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## 2.2 Research Gap

The literature shows that coding agents can now make repository-scale changes [2], [3], [4], that their code and actions are risky in measurable ways [5], [6], [7], [8], that container security depends on configuration [11], [12], [13], and that governance research recommends human approval, legible logs and monitoring [9], [10]. Agent vendors have since shipped security features of their own: OS-level sandboxes for the commands an agent runs in Claude Code [16] and the OpenAI Codex CLI [18], a firewalled hosted environment for the GitHub Copilot cloud agent [19], local microVM sandboxes in Docker Sandboxes [20], and a Docker runtime with an event stream in OpenHands [21]. Table 2.2 compares them with VaultShield, based on their official documentation.

**Table 2.2:** Comparison with existing AI coding-agent security solutions

| Control | Claude Code [16] | Codex CLI [18] | Copilot cloud agent [19] | Docker Sandboxes [20] | OpenHands [21] | VaultShield |
|---|---|---|---|---|---|---|
| Agent compatibility | Claude Code only | Codex only | Copilot only | Several agents | Its own agents | Any agent with an adapter (four built) |
| Isolation approach | OS sandbox for shell commands | OS sandbox | Hosted ephemeral environment | Local microVM | Docker container | Docker container per run (shared kernel) |
| Real .env kept from agent | – | – | Only copilot-environment secrets | – | – | Shape-preserving dummy values |
| Host checkout protected | – | – | Yes (hosted) | – | Configurable | Fresh clone; host never mounted |
| Network restricted by default | Configurable | Yes | Yes (firewall) | Configurable | – | Opt-in allowlist proxy |
| Monitoring and audit evidence | – | – | Session logs | – | Event stream | Event log, egress log, screenshots |
| Change record outside agent's control | – | – | – | – | – | Separate capture helper |
| Human approval before repository submission | Per-action prompts | Per-action approval policy | Review after the branch is pushed | – | Per-action confirmation | Whole change held until approved |

"–" means the cited documentation does not describe the control. The comparison exposes four gaps. First, containment is tied to a vendor: each sandbox protects only its own agent, so a team using several agents has no single policy or evidence format. Second, the project's own .env sits inside the workspace that local sandboxes allow, and deny rules proved brittle in our spike (Section 7.6). Third, any activity record is produced beside the agent, so a hostile agent that tampers with Git state could misreport its change. Fourth, approval is either per action or comes after the change has already been pushed to the repository host. Existing products are ahead of VaultShield on isolation strength and default-deny networking. VaultShield's contribution is to close the four gaps together for any agent on the developer's own machine: dummy credentials and a one-shot clone token keep real secrets away, a separate trusted capture helper builds the change record, and an approval state machine holds the whole change before anything is pushed.

# Chapter 3

# Project Design

## 3.1 Existing System

Developers currently run AI coding agents in one of three ways. The most common is raw local execution, where the agent's CLI or editor extension runs in the developer's own checkout with the developer's environment, permissions and network access, and its changes appear as uncommitted edits. Some products add permission prompts or per-tool deny rules, but these are product-specific and often relaxed by users who find repeated prompts tedious. The second practice is ad-hoc containerisation, which usually bind-mounts the real working tree, so every write still lands on the host and the real .env file travels into the container. The third, development containers and cloud workspaces, is designed for convenience rather than for constraining an untrusted actor and holds nothing for review. Figure 3.1 summarises this workflow.

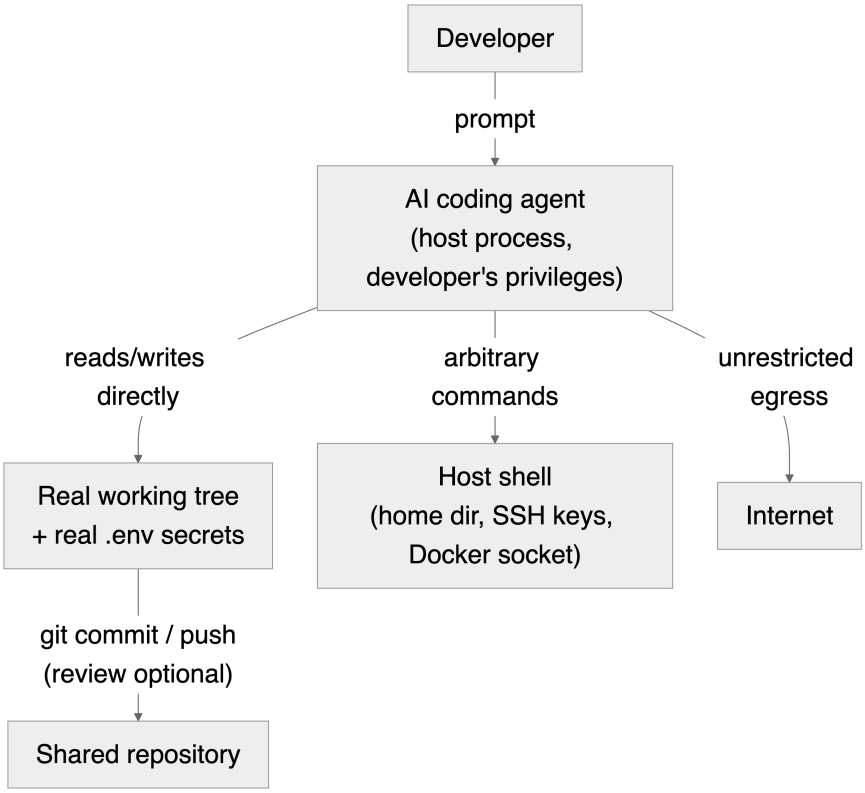



**Figure 3.1:** Existing workflow – direct agent execution on the developer's host

As Figure 3.1 shows, execution and acceptance are a single step and every arrow out of the agent touches a real asset. There is no isolation of side effects, no separation between the agent and the developer's secrets, no structured record of what happened, no enforced review before changes are committed, no automatic cleanup of processes the agent started,

### 3.4.4 Sequence Diagram

Figure 3.9 shows the message exchange for one approved run. Messages 1 to 9 cover the run, 10 and 11 the review, and 12 to 16 the submission. The backend learns of the run before Docker creates anything, the branch and diff are produced by a different container from the one the agent controlled, and submit re-reads the status immediately before pushing.

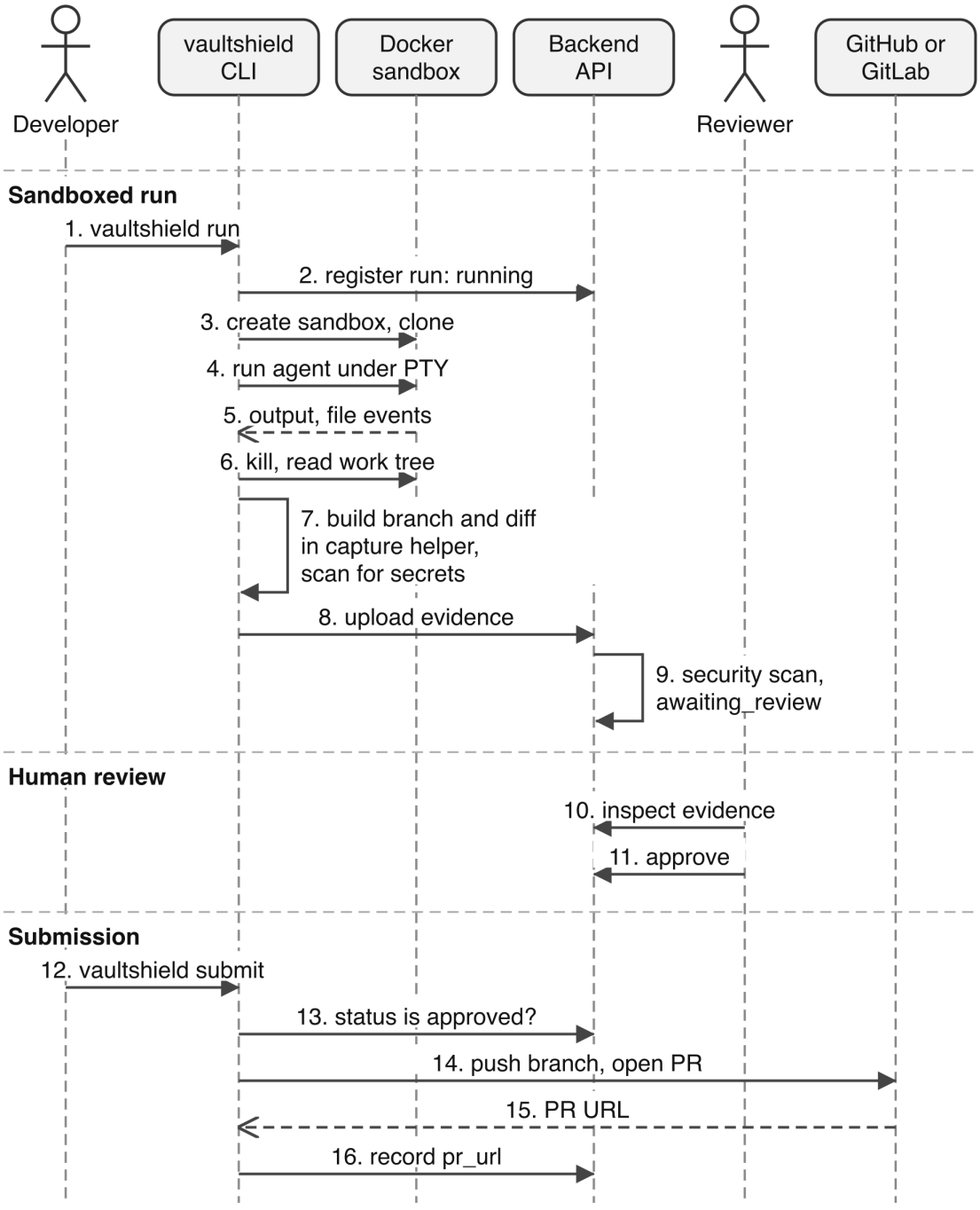



**Figure 3.9:** Sequence diagram of one complete approved run

### 3.4.5 Proposed Agentic AI Orchestrator Workflow

The Agentic AI Orchestrator is Objective 2. It is a proposed design: it has **not** been implemented in Semester VII, is not part of the system evaluated in this report, and is planned for Semester VIII. This subsection specifies how it is intended to work so that its design can be reviewed before implementation.

**Purpose.** Coding agents often receive a one-line prompt such as "add a login page" and must guess the framework, the files to change, the constraints and what counts as done. Wrong guesses produce changes that the reviewer rejects, and they invite hallucinated files, APIs and dependencies. The Orchestrator runs before the agent starts. It reads the project's context, asks the developer a few clarifying questions, and turns the answers into one structured prompt that the agent then receives inside the existing sandbox. It does not replace the agent and it never executes code itself.

**Proposed Workflow.** Figure 3.10 shows the proposed workflow in three stages, and the steps below describe it.

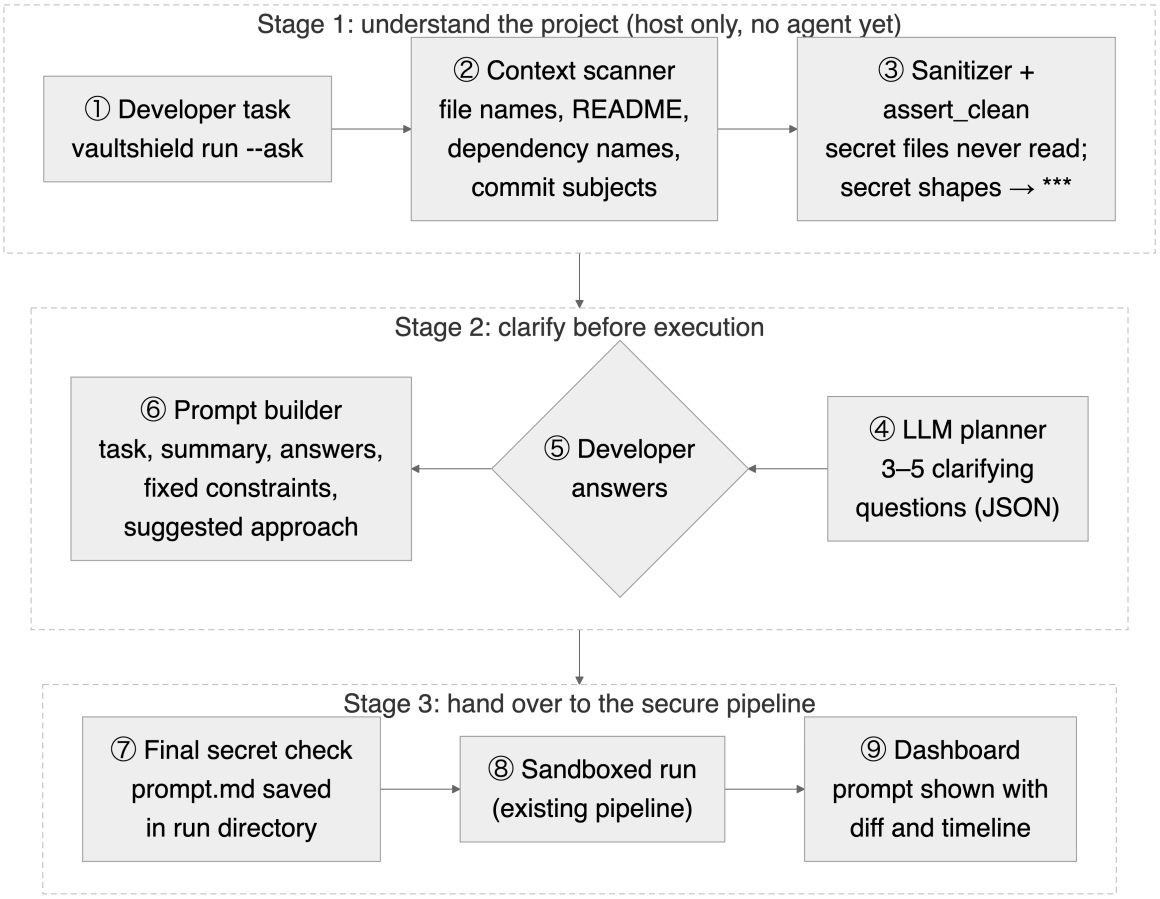



**Figure 3.10:** Proposed workflow of the Agentic AI Orchestrator (Semester VIII)

1. **Task entry.** The developer starts a run with a task description instead of a literal prompt, for example vaultshield run --ask "Add a login page".
2. **Context scanning.** On the host, before any container exists, a scanner collects a bounded project summary: the directory listing to three levels, the start command

and port detected by vaultshield init, dependency names from the package manifests, the beginning of the README and recent commit subjects.

3. **Sanitisation.** Every piece of text passes through a sanitiser that replaces known protected values and secret-shaped strings (API keys, private keys, tokens, passwords in URLs) with ***, followed by an assertion that rejects the request if anything secret-like remains. Secret files such as .env, key files and credential stores are never opened.
4. **Clarifying questions.** A language model receives the sanitised summary and the task and returns three to five clarifying questions in a strict JSON format, for example which authentication method to use or which files must not change. An invalid reply is retried once, and a fixed set of generic questions is used if the model fails.
5. **Developer answers.** The developer answers at the terminal, from a JSON file for scripted use, or skips the questions.
6. **Prompt construction.** The Orchestrator writes an enriched prompt with five parts: Task, Project summary, Answers, Constraints and Suggested approach. The Constraints section is fixed text that the model cannot change, and it tells the agent to refer to secrets by variable name only.
7. **Final check and hand-over.** The finished prompt is checked for secrets again, saved as prompt.md in the run directory and passed to the unchanged sandboxed pipeline described in Section 3.2.
8. **Review.** The dashboard shows the prompt next to the diff and timeline, so the reviewer can judge the change against what was actually asked.

**Proposed Components and Expected Functionality.** Table 3.2 lists the planned components and the function each is expected to provide.

**Table 3.2:** Planned components of the Agentic AI Orchestrator

| Component | Expected functionality |
|---|---|
| Context scanner | Reads a fixed list of non-secret files and produces a size-bounded project summary; refuses symbolic links so a README cannot point at a secret file |
| Sanitiser and leak check | Replaces protected values and secret-shaped text with ***, then blocks the model request if anything secret-like remains |
| Model provider | Sends the sanitised request to a configurable language model with the API key in a header, a 30-second timeout and provider-side storage disabled where the API allows it |
| Question generator | Produces three to five validated clarifying questions, with one retry and a generic fallback |
| Prompt builder | Assembles the five-part enriched prompt with a fixed Constraints section |
| Call log | Appends one metadata line per model call (time, provider, model, request size, outcome) to orchestrator.jsonl, with no request or reply content |
| Pipeline integration | Starts the existing sandboxed run with the enriched prompt and stores prompt.md as part of the run evidence |

**Security Constraints and Evaluation Plan.** The Orchestrator sends project context to an external model, so it is itself a new data-leak path and it must not weaken the guarantees of the Semester VII pipeline. Three constraints follow. First, no secret file is read and every outgoing request passes the leak check, so a failure blocks the request

instead of sending it. Second, the Orchestrator runs before the run is registered and outside the sandbox, but it gains no new privileges: the agent still receives only dummy credentials and its change still goes through trusted capture and human approval. Third, a README containing hostile instructions could steer the model's questions or suggested approach, so the generated prompt is shown to the developer and stored as evidence for the reviewer rather than trusted blindly. Sanitisation by pattern is a heuristic; a secret in an unusual shape inside README text or a commit message may not be recognised, and this residual risk will be stated in the Semester VIII evaluation.

The Semester VIII evaluation will compare runs of the same real agents on the same tasks with a plain prompt and with an Orchestrator prompt, measuring the task success rate against each repository's tests, the reviewer approval rate, the number of hallucinated files, functions or packages in the diff, and the extra time and model cost the Orchestrator adds per run. A separate leak test will seed repositories with known secrets in README files, commit messages and file names and count how many reach the model request.

### 3.4.6 Threat Model and Attack Paths

The threat model states what VaultShield protects, from whom, where the trust boundaries lie and which attacks it addresses. It follows the STRIDE categories (Spoofing, Tampering, Repudiation, Information disclosure, Denial of service and Elevation of privilege). Every mitigation in this subsection is implemented and tested in Semester VII; improvements that are only planned for Semester VIII are listed as residual risk, never as a mitigation.

**Assets, Adversary and Assumptions.** The protected assets are the developer's real secrets (the application .env, SSH keys, cloud credentials and the repository-host token), the developer's working tree and host system, the shared repository and its protected branches, the integrity of the evidence shown to the reviewer, and the agent's own model credential.

The adversary is the AI coding agent and anything that can influence it. The agent may simply be wrong, it may be steered by indirect prompt injection planted in the repository's README, issues or dependencies [7], or it may be deliberately hostile, for example a compromised agent binary or a malicious package that runs during dependency installation. The adversary controls every process inside the sandbox and can read and write every file there.

The following are assumed trusted: the developer's host operating system and Docker daemon, the VaultShield CLI and its capture helper image, the backend operator, and the repository host. In the single-user Semester VII system, the developer and the reviewer are the same trust domain.

**Trust Boundaries.** Figure 3.11 shows the three zones. The untrusted zone contains the agent and its inputs. The sandbox boundary is a per-run container that holds only a fresh clone, a read-only dummy credential file and, when egress control is enabled, a route to the filtering proxy. The trusted zone contains the CLI, the capture helper, the backend and the reviewer, and it is the only place real secrets exist. Labels T1 to T5 mark the main attack paths that cross a boundary.

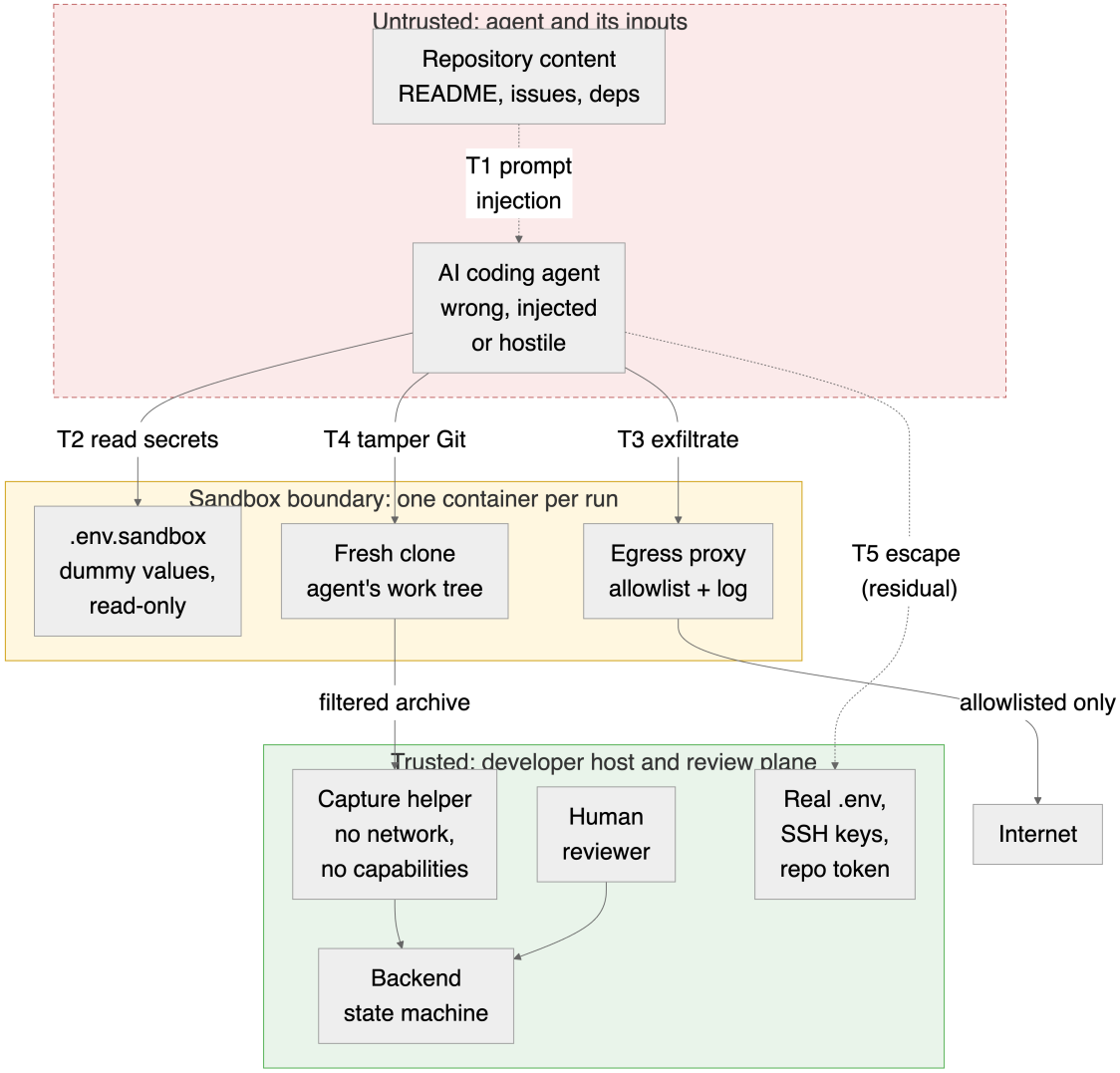



**Figure 3.11:** Threat model: trust zones and attack paths

**Attacks and Mitigations.** Table 3.3 lists the attacks considered, the VaultShield control for each, the evidence that the control works and the risk that remains.

**Table 3.3:** Threat model: attacks, mitigations and residual risk

| ID | Attack | STRIDE | Semester VII mitigation | Evidence | Residual risk and Semester VIII plan |
|---|---|---|---|---|---|
| T1 | Prompt injection from repository content makes the agent act against the developer | T, I | Not prevented; its effects are contained by T2 to T9 and every change requires human approval | Design property | Hostile intent inside an approved change must be caught by the reviewer |
| T2 | Agent reads the real .env, SSH keys or home directory | I | Fresh clone inside the container; only a read-only dummy .env.sandbox is mounted; host paths never mounted | TC-01, TC-02, Fig. 7.8 | Values the developer pastes into the prompt or repository |

| ID | Attack | STRIDE | Semester VII mitigation | Evidence | Residual risk and Semester VIII plan |
|---|---|---|---|---|---|
| T3 | Agent exfiltrates code or data over HTTP, raw TCP or DNS | I | Optional egress allowlist: internal network with no gateway, filtering proxy, public-address check, every decision logged | TC-03, 4 of 4 blocked | Egress is off unless enabled; uploads to an allowlisted host |
| T4 | Repository-host token or the agent's own model key leaks into the diff, logs or files | I | Token given only to the git clone process; token-hygiene scan fails the run and quarantines the diff | TC-06, 5 of 5 vectors | Deliberately encoded leaks |
| T5 | Agent tampers with Git state, hooks or configuration to hide or misreport its change | T, R | Trusted base recorded before the agent runs; separate capture helper with no network rebuilds the branch; unsafe Git configuration refused; hooks disabled | TC-05, 2 of 2 | None known for this path |
| T6 | Agent edits protected paths such as CI or infrastructure files | T | Off-limits paths recorded as violations and the run failed | TC-04 | Paths the developer did not list |
| T7 | Agent damages the host or leaves processes running | T, D | Agent runs only through exec in a disposable container; teardown on every exit path; scheduled reaper | Host unchanged 40 of 40; 0 leftovers | Resource exhaustion inside the sandbox (no default limits) |
| T8 | Container escape through the Docker socket, a kernel bug or a privileged mount | E | Docker socket never mounted; one file mounted; capture helper drops all capabilities | Probe, Fig. 7.8 | Shared kernel and root inside the default sandbox; addressed in Semester VIII |
| T9 | Unapproved change reaches the repository, or two reviewers race | E, R | Approval state machine in a MongoDB transaction; HTTP 409 for the loser; submit re-checks approval before any push | TC-07, TC-08, Fig. 7.7 | Status PATCH route and a single shared identity; role-based approval in Semester VIII |
| T10 | Evidence uploaded to the backend is altered or misattributed | T, S | Token-authenticated API; run identifier binds sandbox, branch and record; branch bundle has a SHA-256 | Backend tests | A holder of the API token can upload; no per-user signing |

The model makes the boundary of VaultShield's claims explicit. It contains a misbehaving agent and produces evidence the agent cannot falsify, but it does not judge intent: a malicious change that looks correct can still be approved by a careless reviewer. Container isolation, default-open egress and the single-user approval model are the three known weaknesses, and they map directly to the Semester VIII enhancements.

# Chapter 4

# Project Planning and Cost Estimation

## 4.1 Project Development Methodology

VaultShield was developed with an iterative and incremental methodology based on specification-driven development. The team first wrote a requirements specification in which every requirement carried numbered acceptance criteria, and a design document with the architecture, data model and numbered correctness properties. The design was split into small, testable tasks, and each short iteration implemented a slice of tasks together with unit and property-based tests and, where isolation was involved, a Docker-backed integration test. The major phases were:

1. **Problem Study and Feasibility Spike:** survey the literature and measure how a production agent handles .env secrets under four protection configurations.
2. **Requirements and System Design:** write numbered requirements and define the architecture, run lifecycle, data model, REST API and threat model.
3. **CLI Core and Evidence Capture:** implement configuration, credential shaping, the sandbox lifecycle, agent adapters, monitoring, screenshots and Git capture.
4. **Backend and Dashboard:** implement the models, REST API, authentication, security scan, scheduled tasks and the review dashboard.
5. **Submission and Hardening:** implement PR and MR submission and teardown, then add the egress proxy, the trusted capture helper and token hygiene.
6. **Migration and Protect Module:** move the backend to MongoDB with transactional approval, and add the Protect module.
7. **Evaluation and Documentation:** run the tests and performance benchmark and write this report.

## 4.2 Project Schedule and Gantt Chart

The Semester VII activities follow the project Gantt chart, reproduced as Figure 5.16, and the Semester VIII activities are planned extensions. Table 4.1 lists each activity with its duration and target period.

In Semester VIII the project continues with four planned enhancements, none of which is part of the Semester VII system. The first is the Agentic AI Orchestrator of Objective 2, which will scan the project, ask the developer clarifying questions and pass a sanitised, enriched prompt to the existing sandboxed pipeline (Section 3.4.5). The second is stronger sandbox isolation through a hardened default container and an optional gVisor or microVM runtime. The third is improved monitoring with an event-driven file watcher and a cross-check of the event log against the trusted diff. The fourth is role-based approval, which separates developer and reviewer roles so that approval becomes a separation-of-duties control. These enhancements follow directly from the limitations measured in Chapters 6 and 7.

**Table 4.1:** Planned project schedule for VaultShield

| Sr. | Project activity | Duration | Target period |
|---|---|---|---|
| **Semester VII** | | | |
| 1 | Group formation and topic finalisation; identifying the scope and objectives of the project | 1 week | 10 Jul – 17 Jul 2026 |
| 2 | Discussing the project topic with the help of a paper prototype | 1 week | 17 Jul – 24 Jul 2026 |
| 3 | Identifying the functionalities of the project as per the objectives identified | 1 week | 24 Jul – 31 Jul 2026 |
| 4 | Literature review and research paper study | 1 week | 31 Jul – 7 Aug 2026 |
| 5 | Designing the graphical user interface | 3 weeks | 7 Aug – 28 Aug 2026 |
| 6 | Presentation I | 1 day | 4 Sep 2026 |
| 7 | Policy logic development | 1 week | 18 Sep – 25 Sep 2026 |
| 8 | Core model development and training | 1 week | 25 Sep – 1 Oct 2026 |
| 9 | Presentation II | 1 day | 9 Oct 2026 |
| 10 | Incorporating the suggestions from Presentation II and submitting a draft copy to the project guide for proofreading | 1 week | 16 Oct – 23 Oct 2026 |
| 11 | Draft submission to the guide for review | 2 days | 23 Oct – 24 Oct 2026 |
| 12 | Final black book submission | 1 week | 26 Oct – 30 Oct 2026 |
| 13 | External presentation and viva voce | 1 day | 2 Nov 2026 |
| **Semester VIII** | | | |
| 14 | Implementation of the Agentic AI Orchestrator (Objective 2) | 3 weeks | January 2027 |
| 15 | Stronger sandbox isolation, improved monitoring and fixes for issues found in testing | 3 weeks | January – February 2027 |
| 16 | Verification of the Codex and Kiro egress allowlists; role-based approval with separate reviewer identities | 2 weeks | February 2027 |
| 17 | Evaluation with real agents on sample repositories, including the Orchestrator | 3 weeks | February – March 2027 |
| 18 | Usability study with reviewers and dashboard refinements | 2 weeks | March 2027 |
| 19 | Research paper drafting and submission | 2 weeks | March 2027 |
| 20 | Final report, demonstration and viva preparation | 3 weeks | March – April 2027 |

## 4.3 Team Roles and Responsibilities

The team of four divided responsibilities by architectural plane, with shared ownership of testing and documentation, as recorded in Table 4.2.

**Table 4.2:** Team roles and responsibilities

| Member | Primary role | Responsibilities |
|---|---|---|
| Aayush Subhash Balip | Backend and data engineering | Django REST API, MongoDB models, run state machine, artifact ingest, Celery tasks, Docker Compose stack |
| Krish Mahesh Patil | Frontend and user experience | React dashboard, run detail tabs, decision controls, accessibility and responsive layout |
| Muhammad Anis Momin | Security research and sandbox engineering | Sandbox lifecycle, credential shaping, egress proxy, trusted capture, token hygiene, threat model |
| Sameer Hasan Saiyed | Integration, testing and documentation | CLI pipeline, agent adapters, submit and teardown, test strategy, benchmark and report |

## 4.4 Software Size Estimation

### 4.4.1 KLOC-Based Estimation

The size of the delivered software was measured directly from the repository as logical source lines, excluding blank lines, comments, docstrings, migrations, lock files and generated content. Table 4.3 reports the counts.

**Table 4.3:** Measured software size by component

| Component | Product code, SLOC | Test code, SLOC |
|---|---:|---:|
| CLI package and tools | 12,404 | 16,697 |
| Backend | 1,150 | 2,552 |
| Frontend | 4,356 | 2,511 |
| Deployment configuration | 160 | – |
| **Total delivered product** | **18,070** | **21,760** |

The delivered product therefore measures KLOC = 18.07. Test code is excluded from the primary estimate, because COCOMO sizes delivered source instructions.

## 4.5 Effort Estimation Using COCOMO

### 4.5.1 Basic COCOMO Model

The Basic Constructive Cost Model [15] estimates effort and schedule from size alone. VaultShield was classified as an organic project, built by a small team in a familiar environment with flexible requirements, so the coefficients are a = 2.4, b = 1.05, c = 2.5 and d = 0.38. E is effort in person-months, T is development time in months and P is the average number of persons.

E = a × (KLOC)^b = 2.4 × (KLOC)^1.05
T = c × (E)^d = 2.5 × (E)^0.38
P = E / T

### 4.5.2 Calculation

With KLOC = 18.07, (18.07)^1.05 = 20.884, so E = 2.4 × 20.884 = 50.12 person-months. Then (50.12)^0.38 = 4.426, so T = 2.5 × 4.426 = 11.06 months, and P = 50.12 / 11.06 = 4.53 persons. Table 4.4 summarises the estimate.

**Table 4.4:** Basic COCOMO estimate

| Quantity | Formula | Value for KLOC = 18.07 |
|---|---|---:|
| Effort E | 2.4 × KLOC^1.05 | 50.12 person-months |
| Development time T | 2.5 × E^0.38 | 11.06 months |
| Average staffing P | E / T | 4.53 persons |

The estimate of about 4.5 persons over about 11 months agrees closely with the actual four-member team working across two semesters, which supports the organic classification.

## 4.6 Human Resource Cost

The project was completed by the four team members as part of their academic programme, and no salary, stipend or consultant fee was paid. Table 4.5 therefore records actual cash expenditure rather than a hypothetical value for unpaid effort.

**Table 4.5:** Actual human-resource cash expenditure

| **Cost item** | **Actual basis** | **Actual cost in ₹** |
|---|---|---:|
| Student development effort | Unpaid academic project work | 0 |
| Guide supervision and review | No project-specific payment | 0 |
| **Total** | | **0** |

## 4.7 Hardware, Software and Cloud Infrastructure Cost

The work used existing team laptops, open-source software and local infrastructure, and no paid cloud environment was commissioned, as shown in Table 4.6.

**Table 4.6:** Actual hardware, software and infrastructure cash expenditure

| **Category** | **Actual basis** | **Cost in ₹** |
|---|---|---|
| Hardware | Existing team laptops; no new equipment | 0 |
| Software | Python, Django, React, Git, Docker and test frameworks, all open source | 0 |
| Database and hosting | Local MongoDB replica set and free-plan repository hosting | 0 |
| Cloud and model APIs | No paid deployment or project-funded usage charge | 0 |
| **Total** | | **0** |

## 4.8 Testing, Deployment and Maintenance Cost

Testing and the MVP deployment ran on local machines with Docker, so no testing or deployment cost was incurred, as shown in Table 4.7. Future maintenance is not included because it has not been incurred.

**Table 4.7:** Actual testing, deployment and maintenance cash expenditure

| **Item** | **Actual basis** | **Actual cost in ₹** |
|---|---|---|
| Automated and Docker-backed tests | Local machines and open-source frameworks | 0 |
| MVP deployment | Local Docker Compose stack | 0 |
| **Total** | | **0** |

## 4.9 Total Estimated Project Cost

Table 4.8 consolidates the actual project-specific cash expenditure.

**Table 4.8:** Total actual project cash expenditure

| **Cost head** | **Source** | **Actual cost in ₹** |
|---|---|---|
| Human resources | Table 4.5 | 0 |
| Hardware, software and infrastructure | Table 4.6 | 0 |
| Testing, deployment and maintenance | Table 4.7 | 0 |
| **Total actual project cash expenditure** | | **0** |

## 4.10 Assumptions and Cost Justification

Only amounts paid specifically for the project are included. Unpaid student effort, depreciation of equipment already owned, normal utilities and possible future maintenance are not given hypothetical monetary values, and tools already available at no extra charge count as zero. The actual cash expenditure for VaultShield is therefore ₹0. This basis is separate from the COCOMO estimate, which describes effort and schedule rather than money spent.

## 4.11 Risk Assessment and Mitigation

Table 4.9 lists the main technical, security and project risks with their likelihood, impact and mitigation.

**Table 4.9:** Risk assessment and mitigation

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Container escape through a kernel or Docker vulnerability | Low | Very high | Docker socket never mounted; only a dummy credential file is mounted; gVisor or microVM isolation planned |
| Malicious dependency running install hooks | Medium | High | Installs only inside the disposable sandbox with dummy credentials; egress allowlist available |
| Network exfiltration when egress control is off | Medium | High | Log-only and allowlist modes through a hardened proxy; default-deny egress planned |
| Real secrets reaching the sandbox or artifacts | Low | Very high | Dummy credentials, one-shot clone credential, token-hygiene scan with diff quarantine |
| Agent tampering with Git state to hide changes | Low | High | Trusted capture in a separate helper from a pre-agent base |
| Approval gate bypassed by a token holder | Medium | Medium | Single-developer trust domain documented; reviewer identities planned |
| Upstream agent CLI changes break an adapter | High | Medium | Adapter interface isolates changes; mock adapter keeps the pipeline testable |
| Schedule slippage or member unavailability | Medium | Medium | Working pipeline after every iteration; shared module knowledge through testing |

# Chapter 5

# Project Implementation

## 5.1 Code Snippets

### 5.1.1 Launching the Sandbox Container

The first snippet is the core of Sandbox.create. Before any Docker call it checks that the credential file exists, because Docker silently creates an empty directory at a missing bind-mount source. Egress options are passed only when they are set, and any failure removes the partial container and raises a single SandboxError. As Figure 5.1 shows, the only host path given to the container is the read-only credential file, and teardown force-removes the container from a finally block so that it runs on every exit path.

Sandbox container creation and teardown

```
# Sandbox.create (body, docstring elided)
try:
    if not env_file.is_file():
        raise SandboxError(
            f"Sandbox creation failed for run {self.run_id}: "
            f"credential file not found at {env_file}"
        )

    self._ensure_image(base_image)

    egress_kwargs = {
        key: value
        for key, value in (
            ("network", network),
            ("cap_drop", cap_drop),
            ("sysctls", sysctls),
            ("extra_hosts", extra_hosts),
        )
        if value is not None
    }
    self.container = self._client.containers.create(
        base_image,
        command=_KEEP_ALIVE_CMD,                # ["tail", "-f", "/dev/null"]
        name=container_name(self.run_id),        # "vaultshield-run-<run_id>"
        working_dir=_WORKDIR,
        volumes={
            str(env_file.resolve()): {
                "bind": _ENV_MOUNT_PATH,         # /workspace/.env.sandbox
                "mode": "ro",
            }
        },
        ports=(
            {f"{publish_port}/tcp": None} if publish_port is not None else None
        ),
        **egress_kwargs,
    )
    self.container.start()

    self._clone_repo(repo_url, resolve_credential=resolve_credential)
    self._install_dependencies(project_dir)
except Exception as exc:
    self._cleanup_after_failure()
    if isinstance(exc, SandboxError):
        raise
    raise SandboxError(
        f"Sandbox creation failed for run {self.run_id}: {exc}"
    ) from exc

# Sandbox.teardown (docstring elided): idempotent, kills every exec'd process
def teardown(self) -> None:
    if self.container is None:
        return
    try:
        self.container.remove(v=True, force=True)
    except docker.errors.NotFound:
        pass
    finally:
        self.container = None
```

**Figure 5.1:** Sandbox container creation and teardown

### 5.1.2 One-Shot Credential for Private Clones

The second snippet clones a private repository without giving the agent the repository-host token. The token is placed only in the environment of the single git clone exec, and the command line names environment variables rather than containing their values. As Figure 5.2 shows, the token is therefore absent from the remote URL, the process arguments, the container environment and error messages, and its lifetime is exactly the duration of the clone.

Authenticated clone with a one-shot credential helper

```
# Sandbox._clone_repo (docstring elided)
provider = _clone_credential_provider(repo_url)      # strict github/gitlab
allowlist
credential, unavailable_reason = self._clone_credential(
    provider, resolve_credential
)

if credential is not None:
    from vaultshield.submit import git_credential_args

    assert provider is not None  # implied by credential is not None
    config_args, env = git_credential_args(credential, provider)
    self._exec_or_raise(
        # `config_args` names the two environment variables the helper
        # reads; the token itself is only ever in `env`.
        ["git", *config_args, "clone", repo_url, _REPO_DIR],
        workdir=_WORKDIR,
        step="git clone",
        environment=env,                 # applies to this one exec only
        secrets=(credential.token,),     # redacted from any error message
    )
    return

try:
    self._exec_or_raise(
        ["git", "clone", repo_url, _REPO_DIR],
        workdir=_WORKDIR,
        step="git clone",
        environment={_GIT_TERMINAL_PROMPT_ENVVAR: "0"},
    )
except SandboxError as exc:
    if unavailable_reason is None:
        raise
    raise SandboxError(
        f"Sandbox creation failed for run {self.run_id} during git "
        f"clone: {repo_url} could not be cloned, and no usable "
        f"{provider} credential was available to authenticate it. If "
        f"the repository is private, {unavailable_reason} "
        f"Original git failure: {exc}"
    ) from exc
```

**Figure 5.2:** Authenticated clone with a one-shot credential helper

### 5.1.3 Event Logging and Off-Limits Enforcement

The third snippet, in Figure 5.3, shows the event record type, the off-limits path matcher and the append-only JSON Lines writer. Every record has exactly three fields matching the backend's RunEvent model, and json.dumps keeps each record on one line, so a partially written log stays readable. The monitor raises OffLimitsViolation only after the writer has run.

Event records, off-limits matching and the JSON Lines writer

```
@dataclass(frozen=True)
class EventLogRecord:
    event_type: Literal["output", "file_change", "violation"]
    timestamp: str  # ISO-8601
    payload: str


def _match_off_limits(
    changed_path: str,
    workdir: Path,
    entries: tuple[str, ...],
) -> str | None:
    # (docstring elided) returns the first violated entry, or None
    if not entries:
        return None
    candidates = _path_candidates(changed_path, workdir)
    for entry in entries:
        for candidate in candidates:
            if candidate == entry or candidate.startswith(entry + "/"):
                return entry
            if fnmatch.fnmatch(candidate, entry):
                return entry
    return None


def _append_event_log(path: Path, records: Iterable[EventLogRecord]) -> Path:
    # (docstring elided) append-only: an existing log is never rewritten
    path.parent.mkdir(parents=True, exist_ok=True)
    with path.open("a", encoding="utf-8", newline="\n") as log_file:
        for record in records:
            log_file.write(json.dumps(asdict(record), ensure_ascii=False) + "\n")
    return path
```

**Figure 5.3:** Event records, off-limits matching and the JSON Lines writer

### 5.1.4 Egress Allow/Deny Decision

The fourth snippet, in Figure 5.4, is the egress proxy's decision function, evaluated for every CONNECT request and TLS connection. A host that is not on the allowlist is never resolved in allowlist mode, IP literals must match an IP entry exactly, and every resolved address must be publicly routable, so an allowlisted name cannot be pointed at an internal address. In log-only mode only the allowlist rule is relaxed.

Egress decision for one request target

```
def decide(
    entries: Sequence[AllowEntry],
    mode: str,
    host: str,
    port: int,
    resolver: Resolver = default_resolver,
    *,
    address_policy: Callable[[str], bool] = is_public_address,
) -> Decision:
    norm = normalise_host(host)
    if norm is None or not isinstance(port, int) or not 1 <= port <= 65535:
        return Decision("deny", "bad_request", host if norm is None else norm,
                        port if isinstance(port, int) else 0)

    if _ip_or_none(norm) is not None:
        if not host_listed(entries, norm):
            return Decision("deny", "ip_literal", norm, port)
        if not matches(entries, norm, port):
            return Decision("deny", "port_not_allowed", norm, port)
        if not address_policy(norm):
            return Decision("deny", "private_address", norm, port)
        return Decision("allow", None, norm, port, (norm,))

    verdict = "allow"
    reason: str | None = None
    if not host_listed(entries, norm):
        if mode != "log-only":
            return Decision("deny", "not_in_allowlist", norm, port)
        verdict, reason = "would_deny", "not_in_allowlist"
    elif not matches(entries, norm, port):
        return Decision("deny", "port_not_allowed", norm, port)

    addresses = _resolve_with_timeout(resolver, norm, port)
    if not addresses:
        return Decision(verdict, reason, norm, port, ())
    if not all(address_policy(address) for address in addresses):
        return Decision("deny", "private_address", norm, port)
    ordered = sorted(addresses, key=lambda a: 0 if ":" not in a else 1)
    return Decision(verdict, reason, norm, port, tuple(dict.fromkeys(ordered)))
```

**Figure 5.4:** Egress decision for one request target

### 5.1.5 Branch Commit and Structured Diff

The fifth snippet, in Figure 5.5, runs inside the trusted capture helper. commit_run_branch refuses unsafe repository configuration and creates exactly one commit, using --allow-empty so that a run which changed nothing still produces a reviewable result and --no-verify so that no repository hook runs. build_structured_diff disables rename detection, external diff programs and text conversion filters.

Committing the run branch and computing the diff

```
def commit_run_branch(sandbox: Sandbox, run_id: str) -> str:
    branch = run_branch_name(run_id)              # "vaultshield/run-<run_id>"

    # Refuse a gitfile/symlinked `.git` and any filter/diff driver
    # BEFORE `git add`, which is what would run a filter.
    _assert_capture_repo_is_safe(sandbox)

    _exec_or_raise(sandbox, _git("checkout", "-b", branch),
                   step=f"creating branch {branch}")
    _exec_or_raise(sandbox, _git("add", "--all"),
                   step="staging working-tree changes")
    _exec_or_raise(
        sandbox,
        _git("-c", f"user.name={COMMITTER_NAME}",
             "-c", f"user.email={COMMITTER_EMAIL}",
             "commit", "--allow-empty", "--no-verify",
             "-m", _commit_message(run_id)),
        step=f"committing changes to {branch}",
    )
    return branch


# build_structured_diff (excerpt): per-file counts, then size-capped patches
result = _exec_or_raise(
    sandbox,
    _git("diff", "--numstat", "-z", "--no-renames",
         "--no-ext-diff", "--no-textconv", base_ref, branch),
    step=f"diffing {base_ref}..{branch}",
)
records = _parse_numstat(_raw_stdout(result))
metadata_cost = _metadata_cost(records)
if metadata_cost > MAX_DIFF_REQUEST_BYTES:          # 2 MiB upload budget
    raise GitCaptureError(
        "Git capture failed while sizing diff.json: the entries for "
        f"{len(records)} changed files take {metadata_cost} bytes even "
        f"with every patch left out, over the {MAX_DIFF_REQUEST_BYTES}-byte "
        "upload cap (MAX_DIFF_REQUEST_BYTES), so the backend would reject "
        "the diff"
    )
```

**Figure 5.5:** Committing the run branch and computing the diff

### 5.1.6 Approval State Machine

The sixth snippet, in Figure 5.6, is the inner function of apply_decision. The status check and status write are a single conditional update, and the Approval record is created in the same MongoDB transaction, so only one of two concurrent decisions can match. The loser receives HTTP 409 naming the run's actual status; before the retry wrapper was added, 54 of 72 losing calls in a concurrency test received a server error instead.

Transactional approve/reject decision

```
ALLOWED_DECISIONS_FROM = frozenset({"awaiting_review"})

def apply_decision(run: Run, decision: Decision) -> Run:
    # (docstring elided)
    decided_at = to_mongo_precision(timezone.now())

    def record_decision() -> None:
        with transaction.atomic():          # django_mongodb_backend's atomic()
            recorded_decision = (
                Approval.objects.filter(run=run)
                .values_list("decision", flat=True).first()
            )
            if recorded_decision is not None:
                raise StateTransitionError(
                    _stored_status_of(run), recorded_decision=recorded_decision
                )

            transitioned = Run.objects.filter(
                pk=run.pk, status__in=ALLOWED_DECISIONS_FROM
            ).update(status=decision, updated_at=decided_at)

            if not transitioned:
                raise StateTransitionError(_stored_status_of(run))

            Approval.objects.create(run=run, decision=decision)

    # Retried on MongoDB's TransientTransactionError only. "Measured at 54
    # spurious errors out of 72 losing calls before this was added."
    retry_on_transient_transaction_error(record_decision)

    run.status = decision
    run.updated_at = decided_at
    return run
```

**Figure 5.6:** Transactional approve/reject decision

## 5.2 Steps to Access the System

The system requires Python 3.11 or later, a running Docker daemon, Git, Node.js 18 or later, a MongoDB replica set or Atlas cluster, and Playwright Chromium. The commands assume macOS or Linux.

1. Clone the repository and copy .env.example to .env, setting VAULTSHIELD_ MONGODB_URL to the MongoDB connection string.
2. Start the backend and dashboard with docker compose up –build. The API is served on http://localhost:8000 and the dashboard on http://localhost:5173.
3. Create a user with docker compose exec web python manage.py createsuperuser.
4. Install the CLI:
   cd cli && python3 -m venv .venv && source .venv/bin/activate && pip install -e ”.[dev]” && playwright install chromium
5. Authenticate with vaultshield login –backend-url http://localhost:8000 –username <user>, create a project in the dashboard and note its identifier.
6. In a checkout of the target repository, run vaultshield init to write .vaultshield.yml.

7. Execute a sandboxed run, for example with the mock agent:
   vaultshield run –agent=mock –prompt=”add a file” –repo-url=”$REPO” –project ”$PROJECT_ID”
8. Open the dashboard, select the run, inspect the Diff, Timeline, Screenshots and Security tabs, and choose Approve or Reject.
9. For an approved run, run vaultshield submit <run-id> to open the pull or merge request; for a rejected run, run vaultshield teardown <run-id>.

## 5.3 System Implementation

The dashboard screenshots below were captured by the repository’s layout tool, which serves the built dashboard with fixture data, so they show the real interface with illustrative data.

### 5.3.1 Command-Line Interface

The CLI provides the pipeline commands init, login, run, submit and teardown and the supporting commands protect and exec. During vaultshield run, the terminal prints one line per pipeline milestone, as shown in Figure 5.7, and each run writes a run directory containing the dummy credential file, events.jsonl, diff.json, the branch bundle and the screenshots.

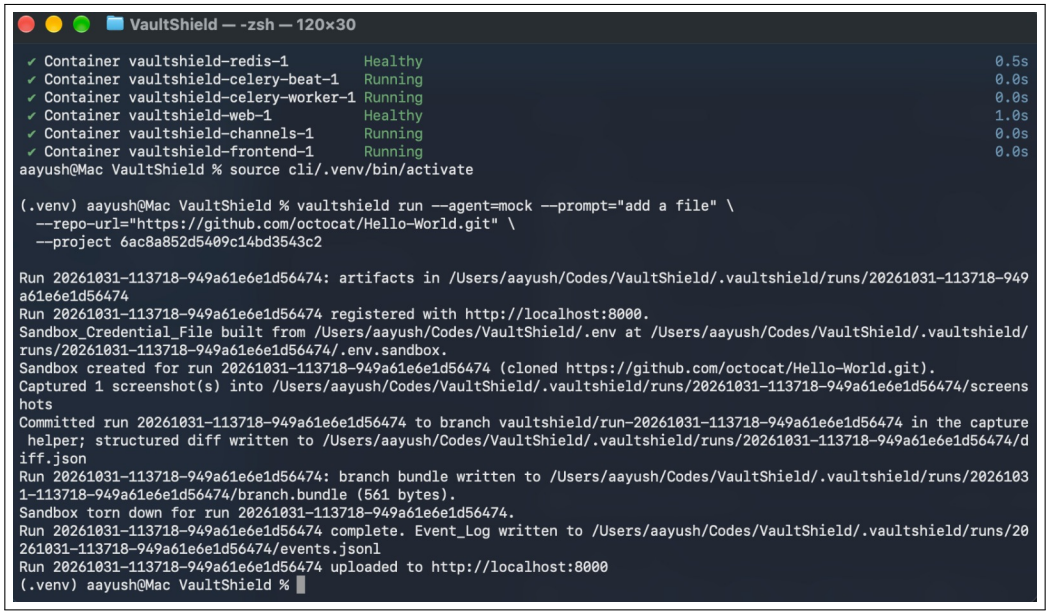



**Figure 5.7:** Terminal output of a successful vaultshield run with the mock agent

### 5.3.2 Landing Page and Projects Dashboard

The landing page in Figure 5.8 introduces the product with a one-line installer command and links to sign in. After sign-in, the projects overview in Figure 5.9 shows one card per project with its repository URL, a sidebar of projects and a New project button.

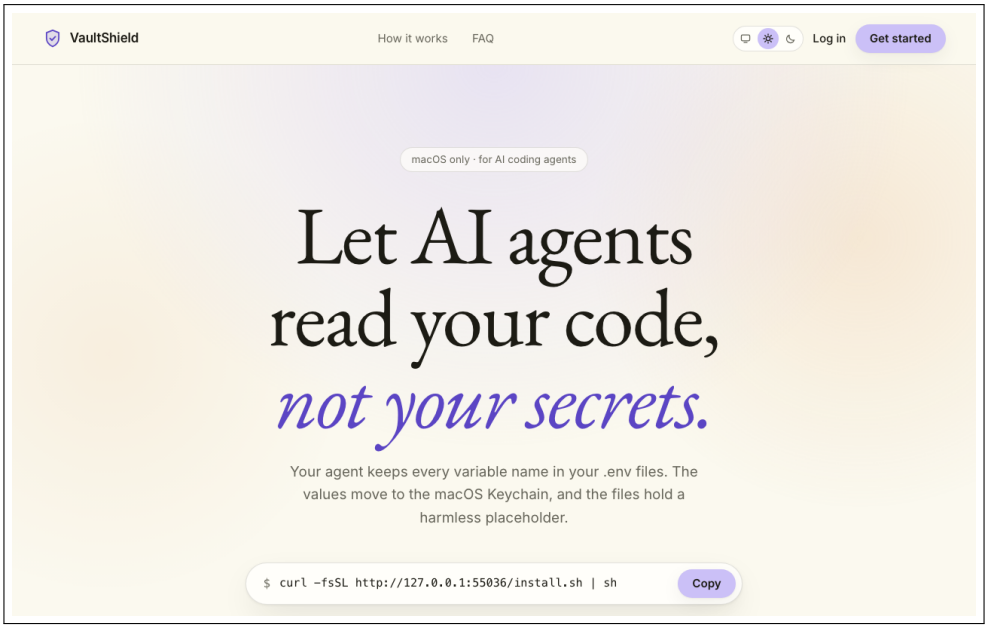



**Figure 5.8:** Landing page

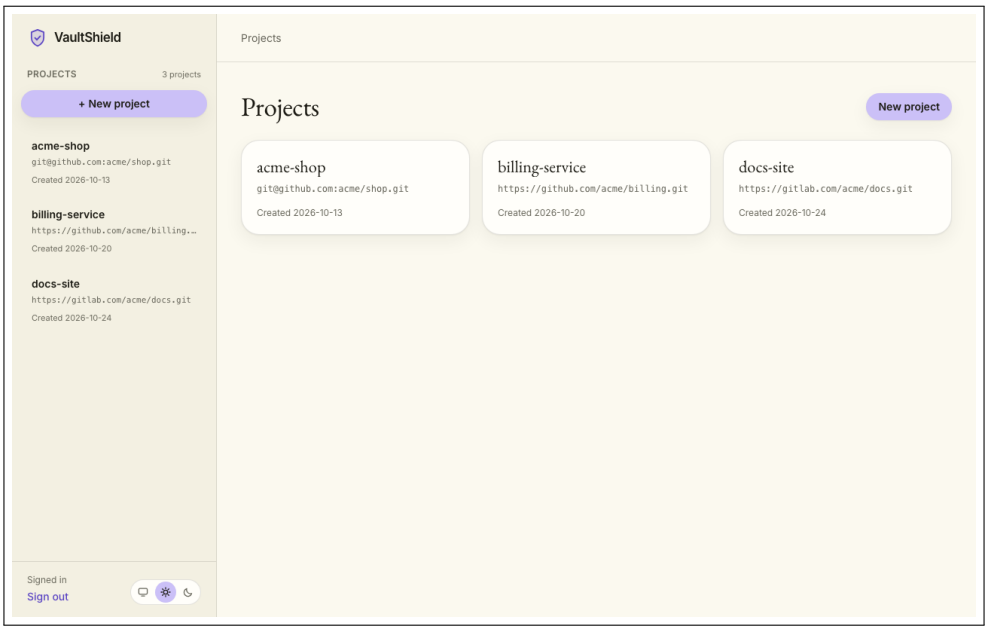



**Figure 5.9:** Projects dashboard

### 5.3.3 Run History

The run history in Figure 5.10 lists a project's runs newest first with the run identifier, agent, status badge and creation time, with status filters, search by run identifier and pagination.

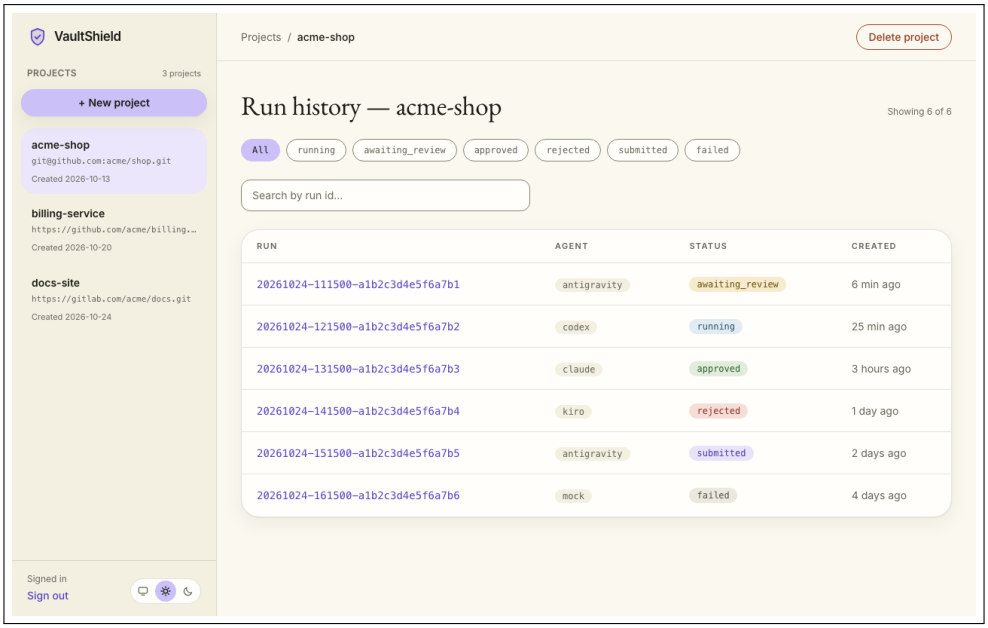



**Figure 5.10:** Run history with status filters

### 5.3.4 Run Detail Views

The run detail page has a header with the run's status, a decision banner while the run is awaiting review, and four tabs. The Diff tab in Figure 5.11 lists every changed file with its addition and deletion counts. The Timeline tab in Figure 5.12 shows the event log in order, giving the reviewer a view of what the agent did as well as what it changed [14]. The Screenshots tab in Figure 5.13 shows one image per configured route, and the Security tab in Figure 5.14 lists the findings of the automated scan with their rule, file and line. Findings inform the reviewer but do not block approval.

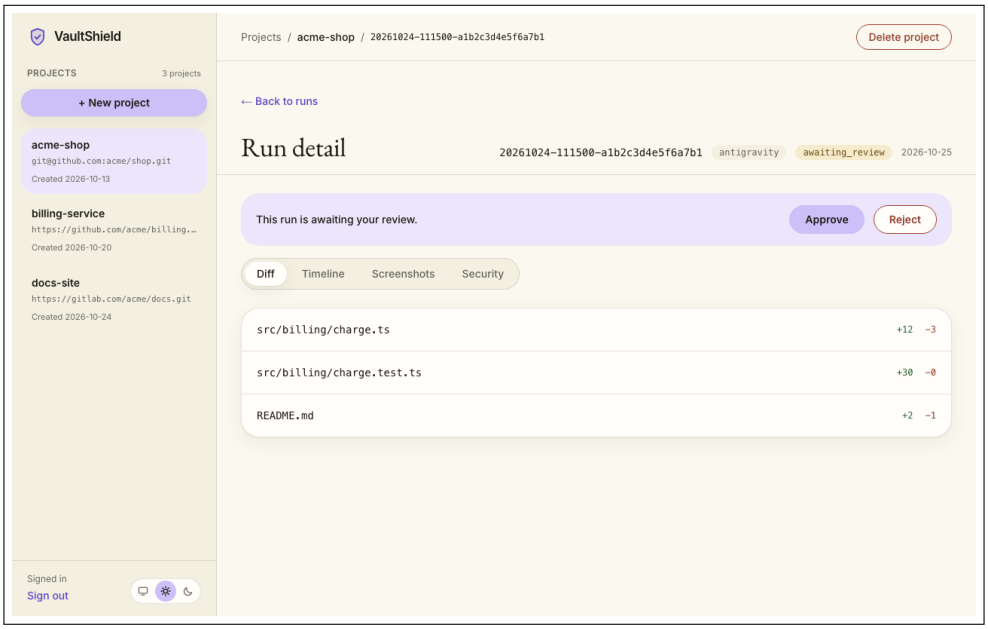



**Figure 5.11:** Run detail – Diff tab

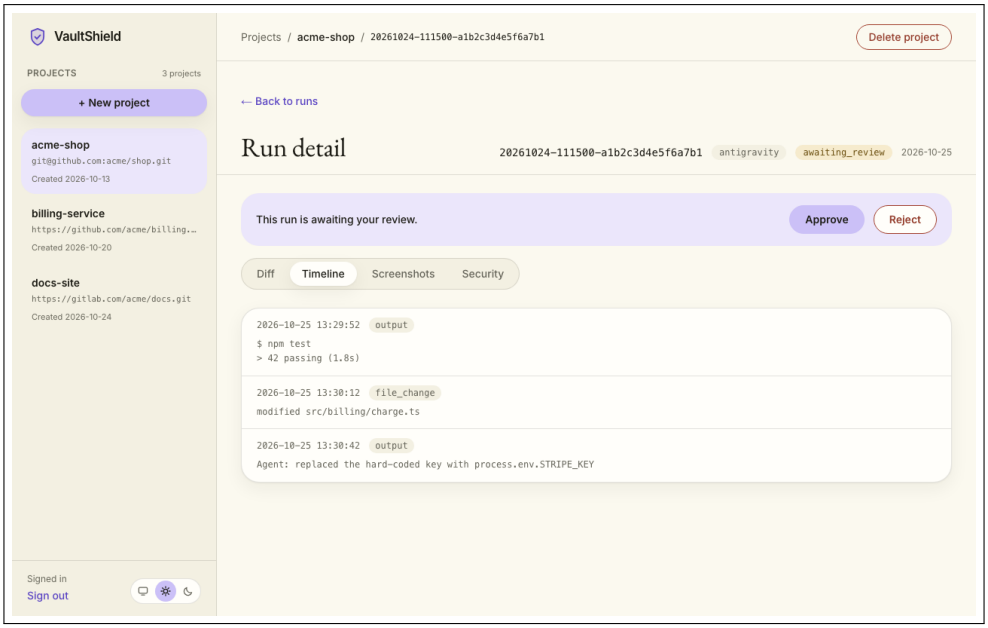



**Figure 5.12:** Run detail – Timeline tab

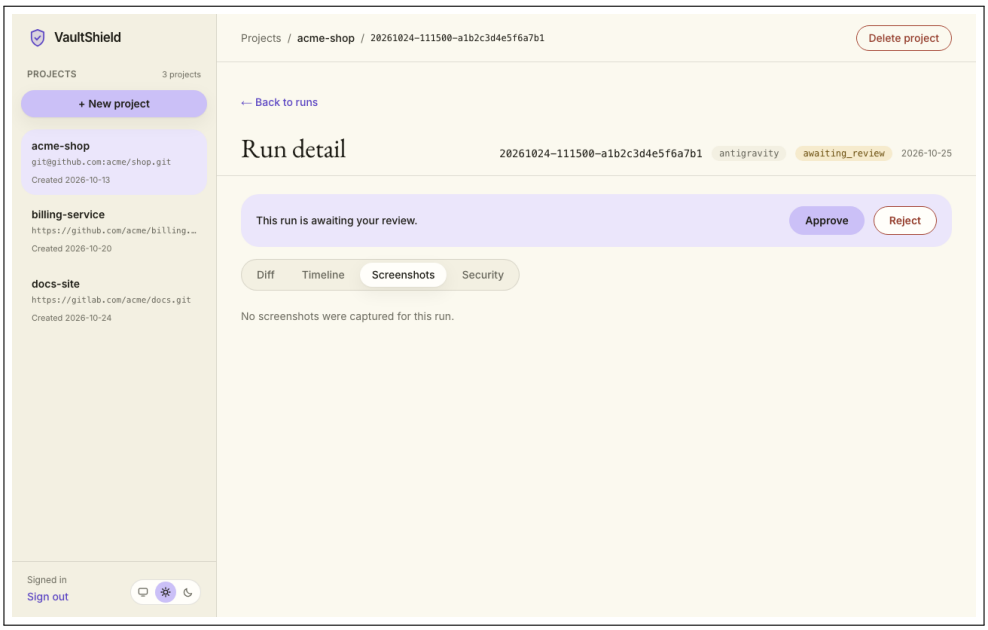



**Figure 5.13:** Run detail – Screenshots tab

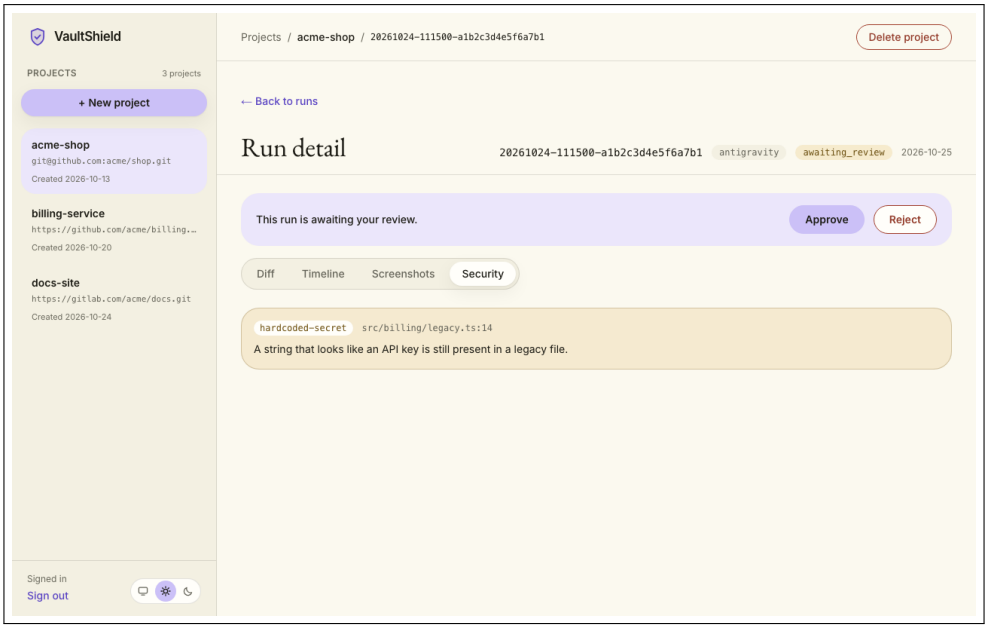



**Figure 5.14:** Run detail – Security tab

### 5.3.5 Approval Decision and Submission

When the reviewer approves a run, the buttons are replaced by the status “approved, awaiting submission” and the exact vaultshield submit command, as shown in Figure 5.15, because the backend never opens a pull request on its own.

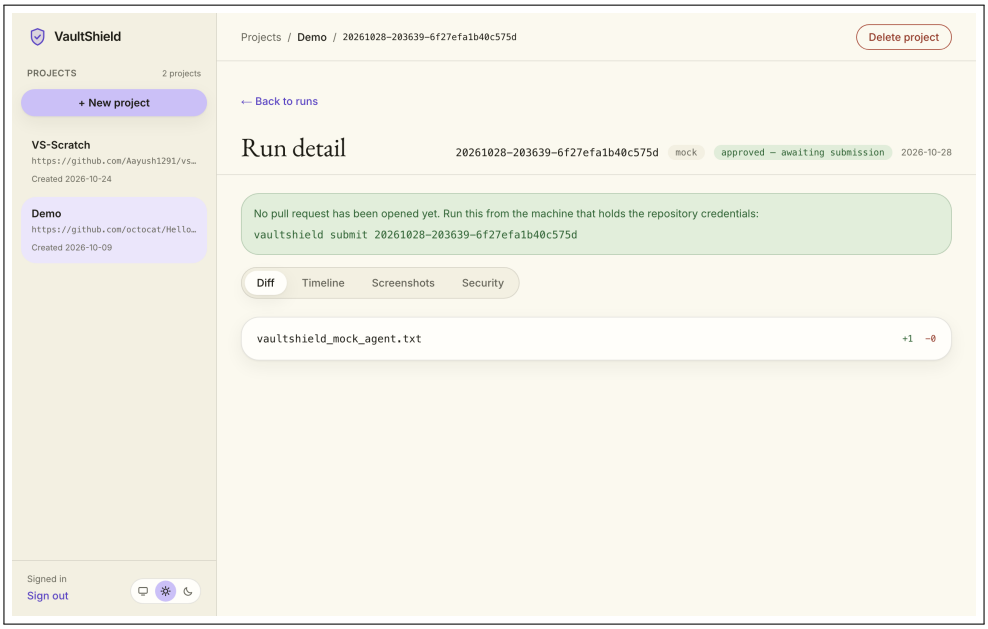



**Figure 5.15:** Run detail after approval, showing the submit command

### 5.3.6 Implementation Summary

The CLI creates the sandbox, monitors the agent, captures the change on a dedicated branch and uploads the evidence, while the dashboard presents that evidence and records the reviewer’s decision. Producing a change and accepting it therefore remain separate

steps. Table 5.1 lists every feature implemented and evaluated in Semester VII, and Table 5.2 lists the enhancements planned for Semester VIII, none of which is part of the delivered system.

**Table 5.1:** Implemented in Semester VII

| Area | Implemented feature |
|---|---|
| CLI and adapters | Python CLI (init, login, run, submit, teardown); mock, codex, kiro and antigravity agent adapters |
| Sandbox execution | Docker-based sandbox per run; fresh repository clone inside the container; guaranteed teardown |
| Credentials | Dummy credential substitution in a read-only .env.sandbox; one-shot repository clone credentials |
| Egress control | Per-run internal network and hardened proxy with log-only and allowlist modes |
| Monitoring | Terminal (PTY) and in-container file monitoring; JSON Lines event logging; off-limits path enforcement |
| Change capture | Trusted change capture in a separate helper; one-commit run branch and structured Git diff |
| Backend and dashboard | Django REST backend with MongoDB integration; React review dashboard |
| Security and approval | Token-hygiene and static security scanning; human approval workflow; approval-gated pull or merge request submission |
| Protect module | Keychain-backed .env protection for agents run outside the sandbox |

**Table 5.2:** Planned for Semester VIII (not implemented)

| Enhancement | Planned work | Semester VII gap addressed |
|---|---|---|
| Agentic AI Orchestrator | Context scanner, sanitiser, clarifying questions and enriched prompt (Section 3.4.5) | Objective 2 not implemented |
| Stronger sandbox isolation | Non-root user, dropped capabilities, resource limits, default-deny egress, gVisor or microVM runtime | Shared kernel; root in the default sandbox; opt-in egress |
| Improved monitoring | Event-driven file watcher, event-log cross-check against the diff, live streaming, sign-in watchdog | Watcher missed 8 of 40 writes; failed test 1 |
| Role-based approval | Reviewer accounts, role-based access control, gated status transitions | Single-user trust domain; status PATCH can set approved |

## 5.4 Timeline Sem VII

The Gantt chart in Figure 5.16 shows the Semester VII activities across thirteen weeks, from group formation on 10 July 2026 to the external presentation and viva voce on 2 November 2026. The activities are grouped into four phases: project conception and initiation, design and implementation, evaluation and documentation, and final submission. All four members are owners of every activity, and every Semester VII activity is complete.

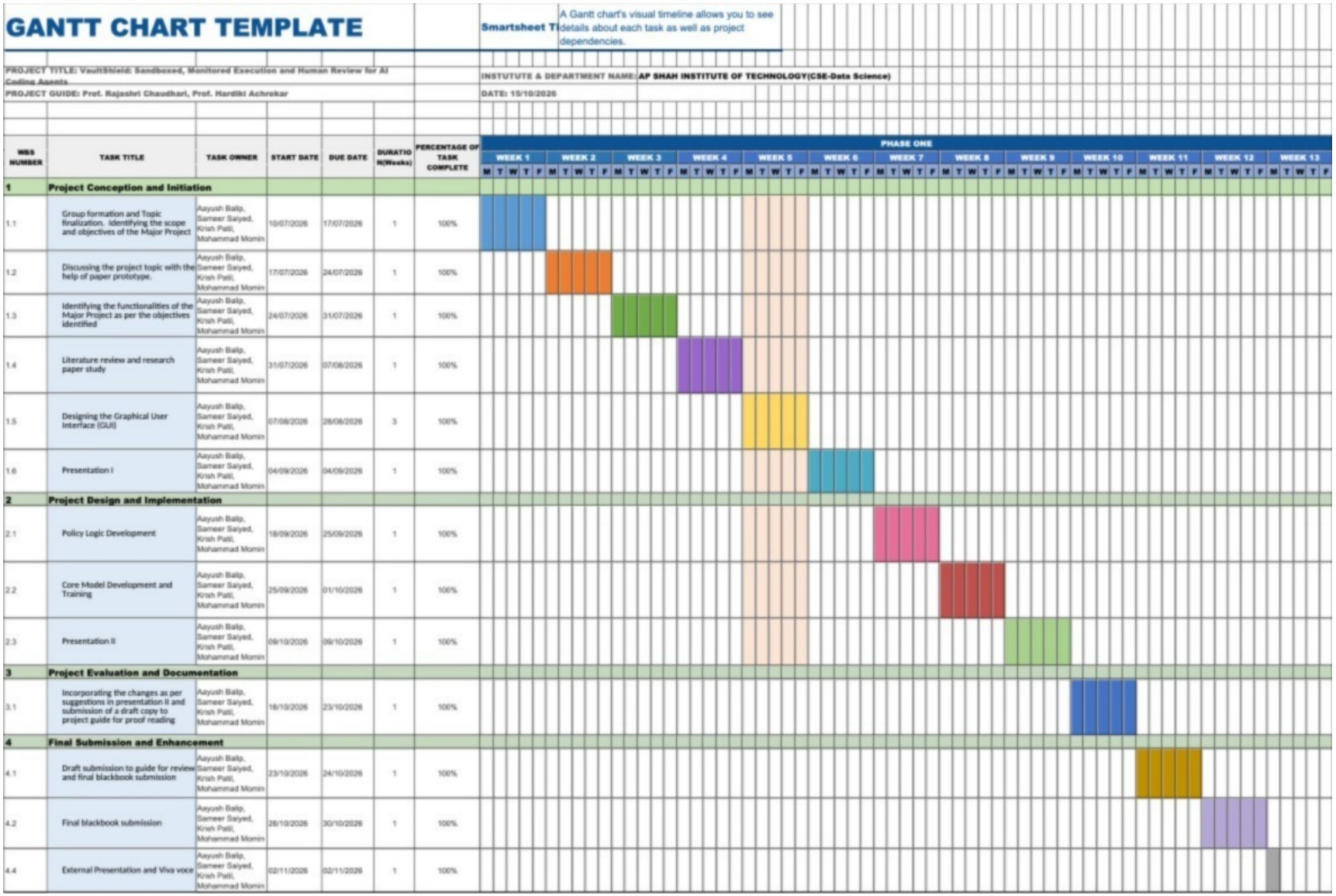



**Figure 5.16:** Gantt chart of the VaultShield major project

# Chapter 6

# Testing

## 6.1 Software Testing

VaultShield's value lies mostly in negative guarantees: the agent must not reach the host, real secrets must not enter the sandbox, a token must not appear in any artifact, and an unapproved run must not become a pull request. The team therefore traced every acceptance criterion to at least one test, expressed invariants as property-based tests with Hypothesis and fast-check, tested isolation claims against real Docker containers with a probing agent rather than mocks, and tested the backend against a real MongoDB replica set because the approval gate depends on genuine transaction semantics. Test fixtures redirect the home directory to an empty temporary directory so that no test can touch the developer's real files.

## 6.2 Types of Testing

Testing was organised into functional testing and system testing. Table 6.1 lists the test levels used within them.

**Table 6.1:** Types of testing performed

| Level | Scope | Tools |
|---|---|---|
| Unit | Configuration parsing, credential shaping, off-limits matching, diff parsing, egress decisions, React components | pytest, Hypothesis, Jest, fast-check |
| Integration | CLI with real Docker, backend with MongoDB, upload with the API | pytest with Docker and a replica set |
| Security-focused | Isolation, secret suppression, network blocking, tamper resistance, approval atomicity | Probing agents in real containers, concurrency tests |
| System | The complete run, submit and teardown flows | Click CliRunner, Docker, timing harness |
| Usability and accessibility | Contrast ratios, responsive layout, keyboard navigation | Jest contrast test, Playwright layout tool |

### 6.2.1 Functional Testing

Functional testing verified that each function, module and security control behaves as specified. Unit tests checked properties such as a dummy credential never containing the real value and the egress decision table for IP literals and private addresses. Security-focused tests checked that a non-allowlisted connection and DNS-based exfiltration are

blocked, that a tampered in-container Git cannot hide a file from the diff, and that concurrent approve and reject requests resolve to exactly one decision.

### 6.2.2 System Testing

System testing drove the real vaultshield run command end to end with a real container, clone, agent process, trusted capture and teardown, and the same harness executed the 40 timed runs reported in Chapter 7.

## 6.3 Testing Results and Findings

Table 6.2 summarises the results of the three automated suites, which form the Semester VII testing campaign.

**Table 6.2:** Automated test suite results

| Suite | Tests | Passed | Failed | Pass rate |
|---|---:|---:|---:|---:|
| CLI | 2,080 | 2,078 | 2 | 99.90% |
| Backend | 341 | 341 | 0 | 100.00% |
| Frontend | 283 | 283 | 0 | 100.00% |
| **Total** | **2,704** | **2,702** | **2** | **99.93%** |

No test was skipped. The lowest text contrast was 4.77:1 in the light theme and 5.80:1 in the dark theme, above the 4.5:1 threshold, and the layout report recorded no horizontal overflow across 63 page and width combinations. Table 6.3 lists the main issues found during development and testing.

**Table 6.3:** Issues found and their resolution

| Issue | Resolution | Status |
|---|---|---|
| 54 of 72 losing concurrent decisions received HTTP 500 instead of 409 | Retry on transient transaction errors | Fixed |
| Dummy connection strings kept the real database host name | Host replaced unless provably local | Fixed |
| A missing credential file became an empty directory through Docker's bind mount | File check before any Docker call | Fixed |
| Host listeners reachable from a plain internal bridge | Internal bridge created without a gateway | Fixed |
| A hanging interactive sign-in was not bounded by its deadline (failed test 1) | Watchdog-based deadline planned | Open |
| A test expected the raw echoed value that redaction now masks (failed test 2) | Test expectation to be updated | Open |
| The file watcher missed the agent's write in some fast runs | The trusted diff remains the authoritative record | Open |

| Issue | Resolution | Status |
|---|---|---|
| The status PATCH route can set approved without an Approval record | Documented MVP limitation; gating planned | Open |

### 6.3.1 Test Environment and Configuration

Table 6.4 records the configuration under which the suites and the benchmark were run.

**Table 6.4:** Test environment and configuration

| Item | Configuration |
|---|---|
| Hardware | Apple MacBook Air, M4 processor |
| Container runtime | Docker Desktop 29.2.1; Docker-backed tests use the vaultshield sandbox image and the real daemon |
| CLI suite | Python, pytest with Hypothesis; command: cd cli && pytest -q |
| Backend suite | pytest against a MongoDB replica set, in a throw-away test_ database; command: cd backend && pytest -q |
| Frontend suite | Jest with React Testing Library and fast-check; command: cd frontend && npm test |
| Isolation of tests | Home directory redirected to an empty temporary directory; containers and networks removed in finally blocks |
| System-test agent | Deterministic mock agent; sample repository https://github.com/octocat/Hello-World.git |

The figures in Table 6.2 are from the Semester VII testing campaign. The suites were re-run on 1 November 2026, as shown in Figure 6.1. The backend (341 of 341) and frontend (283 of 283) results were identical. The CLI suite collected 2,084 tests in the re-run, four more than the 2,080 of the original campaign because tests were added afterwards; 2,082 passed and the same two tests failed. The re-run confirms the campaign result; it does not replace it.

Full automated test suites, re-run on 1 November 2026

```
$ cd cli && pytest -q -rfE   # excerpt: summary lines
FAILED tests/test_sandbox_execution.py::test_exec_interactive_really_delivers_stdin_to_a_container_process
FAILED tests/test_sandbox_execution.py::test_a_hanging_signin_hits_the_read_timeout_instead_of_blocking_forever
2 failed, 2082 passed, 94 warnings in 734.14s (0:12:14)

$ cd backend && pytest -q
341 passed in 421.52s (0:07:01)

$ cd frontend && npm test
Test Suites: 21 passed, 21 total
Tests:       283 passed, 283 total
Snapshots:   0 total
Time:        2.516 s
```

**Figure 6.1:** Re-run of the three automated suites on 1 November 2026 (excerpt of summary lines)

### 6.3.2 Analysis of the Two Failed Tests

Both failing tests are Docker-backed tests in tests/test_sandbox_execution.py. Both exercise the interactive sign-in transport, which is used only when the antigravity agent is run with --interactive-auth. This transport writes a pasted authorisation code to the agent's standard input through the Docker exec socket and redacts it from everything the CLI displays. Figure 6.2 shows both failures re-run individually.




**Figure 6.2:** The two failing tests re-run individually, with their durations (excerpt)

**Failure 1: interactive sign-in timeout.** The test test_a_hanging_signin_hits_the_read_timeout_instead_of_blocking_forever starts a fake sign-in program in a real container that prints the code prompt and then hangs. The test shortens the read deadline to 3 seconds and expects the CLI to give up within 60 seconds. The read deadline is set as a socket timeout on the exec stream, and on Docker Desktop 29.2.1 that timeout does not interrupt the blocking read, so the read blocked for 600.33 seconds until the fake program exited. This is a genuine availability defect in the optional interactive authentication flow: a sign-in that stops responding keeps the CLI waiting instead of failing after the deadline. It exposes no secret, because the code is still redacted and teardown still runs, which other passing tests verify. A watchdog-based deadline that kills the exec independently of the socket is planned.

**Failure 2: redacted echo expectation.** The test test_exec_interactive_really_delivers_stdin_to_a_container_process sends a sentinel value to a container process that prints RECEIVED:[value], and expects the raw value in the output. The actual output was RECEIVED:[***]. Redaction replaces only an exact match of the sent value, so this output proves that the process received exactly the bytes sent and that the terminal echo was then masked. The masking is intentional credential protection that keeps an authorisation code off the terminal and out of the logs. The test predates the redaction, so its expectation is outdated and should be changed to the redacted form.

**Reliability impact.** The two failures are about 0.07% of the 2,704 tests of the original campaign. They are of different kinds. Failure 1 is a genuine functional defect: the code does not do what it is specified to do, and a user of the optional sign-in flow can observe it. Failure 2 is an outdated test assertion: the code behaves as specified and only the test's expectation is wrong, so it is not a functional or security defect. Neither affects the default execution workflow, which does not use the interactive transport. The security-critical isolation, secret-substitution, egress, trusted-capture, token-hygiene and approval tests all passed; they drive the sandbox through the ordinary non-interactive exec path that every default run uses, so a defect in the optional sign-in path cannot hide a failure in them.

## 6.4 Sample Test Cases

Table 6.5 presents representative test cases, each corresponding to an automated test or to the benchmark harness.

**Table 6.5:** Sample test cases

| ID | Description | Expected and actual result | Status |
|---|---|---|---|
| TC-01 | Sandbox isolation | Probing agent cannot read the host .env, home directory or Docker socket; host checkout untouched in 40 of 40 runs | Pass |
| TC-02 | Secret suppression | No generated dummy value contains its real value; 135 of 135 tests passed | Pass |
| TC-03 | Network blocking | CONNECT to unlisted hosts, IP literals and the host answered 403; direct DNS lookups fail | Pass |
| TC-04 | Off-limits enforcement | Write to a protected path records a violation event, then fails the run | Pass |
| TC-05 | Tamper resistance | A file hidden by a tampered in-container Git still appears on the run branch | Pass |
| TC-06 | Token hygiene | All five credential-leak vectors fail the run and quarantine the diff | Pass |
| TC-07 | Approval gating | submit refuses every non-approved run with no push and no PR | Pass |
| TC-08 | Concurrent decisions | Exactly one winner per trial and exactly one Approval record | Pass |
| TC-09 | Monitoring completeness | Every agent write in the timeline; present in 32 of 40 runs, while the diff recorded 40 of 40 | Partial |
| TC-10 | Hanging sign-in is bounded | Expected to stop within 60 s; took 600.3 s | Fail |

## 6.5 Conclusion on Testing

Of 2,704 automated tests in the Semester VII campaign, 2,702 passed, and every security-critical behaviour named in the objectives was exercised against real infrastructure. As analysed in Section 6.3.2, the two failures affect only the optional interactive sign-in transport: one is an availability defect, the other an outdated test assertion. The file watcher's missed events are a gap in supplementary evidence rather than in change capture, because the trusted diff was complete in every run, and the unrestricted status PATCH route is acceptable only within the single-developer trust model of the MVP.

# Chapter 7

# Result and Discussions

## 7.1 Introduction

This chapter evaluates VaultShield with a timing harness that executed 40 complete vaultshield run invocations against real Docker, a separate baseline of 20 runs of the same agent directly on the host, the security-focused test cases of Chapter 6, the feasibility spike on secret exposure, captures of a real execution, and a case study of preserved run directories. The measurements were taken on an Apple MacBook Air with an M4 processor running Docker Desktop 29.2.1, with a small local repository and a stubbed backend.

**Scope of the performance results.** Every timing in this chapter was measured with VaultShield's deterministic *mock agent*, not with a real model-backed agent such as Codex, Kiro or Antigravity. The mock agent runs inside the sandbox through the same adapter, PTY and monitoring path as a real agent, but it makes no model API calls and writes one small file. The results therefore measure the fixed overhead of the VaultShield pipeline under the mock-agent workload. They must not be read as the performance of real agents, whose evaluation is future work planned for Semester VIII.

### 7.1.1 Benchmark Configuration

Table 7.1 lists the four configurations. The benchmark consists of 60 timed executions: 40 VaultShield runs (20 default, 10 with the egress allowlist and 10 with one screenshot route) and 20 direct-host baseline runs, which did not use VaultShield and serve only to compute overhead. Every reference to "the 40 runs" in this report means the 40 VaultShield runs. The raw per-run timing records of this campaign are not preserved in the repository; the counts and figures reported here are those recorded in the benchmark campaign and documented in the project's earlier full report draft, and they have not been independently re-measured.

**Table 7.1:** Benchmark configurations (all with the mock agent)

| Configuration | Runs | Execution | Purpose |
|---|---|---|---|
| Direct-host baseline | 20 | Host clone, no container | Baseline without VaultShield |
| VaultShield default | 20 | Sandbox and trusted capture, egress off | Fixed pipeline overhead |
| VaultShield with egress allowlisting | 10 | As default, egress allowlist on | Cost of network isolation |
| VaultShield with screenshot capture | 10 | As default, one screenshot route | Cost of browser capture |
| **Total** | **60** | **40 VaultShield + 20 baseline** | |

## 7.2 Evaluation Metrics

Table 7.2 defines the metrics used.

**Table 7.2:** Evaluation metrics

| Metric | Definition |
|---|---|
| End-to-end latency | Wall-clock time of one vaultshield run, from command start to exit |
| Stage latency | Time spent in each pipeline stage |
| Overhead | Difference between VaultShield latency and the same agent run directly on the host |
| Isolation effectiveness | Fraction of runs in which the host checkout was unchanged and host files were unreachable |
| Capture and monitoring completeness | Fraction of runs whose diff and event log contained the agent's change |
| Detection of unsafe actions | Fraction of injected unsafe behaviours that were blocked or detected |
| Cleanup reliability | Number of VaultShield containers or networks left after all runs |

## 7.3 Performance Analysis

Table 7.3 reports the stage latency of the default configuration over 20 runs, in seconds, and Figure 7.1 shows the mean of each stage.

**Table 7.3:** Stage latency in the default configuration

| Stage | Mean | Std. dev. | Share |
|---|---:|---:|---:|
| Sandbox creation, clone and install | 0.239 | 0.017 | 17.0% |
| Agent execution and monitoring | 0.413 | 0.033 | 29.4% |
| Trusted capture | 0.651 | 0.053 | 46.4% |
| Token-hygiene scan | 0.019 | 0.008 | 1.4% |
| Teardown | 0.042 | 0.006 | 3.0% |
| Other stages | 0.039 | – | 2.8% |
| **End-to-end total** | **1.403** | **0.075** | **100%** |

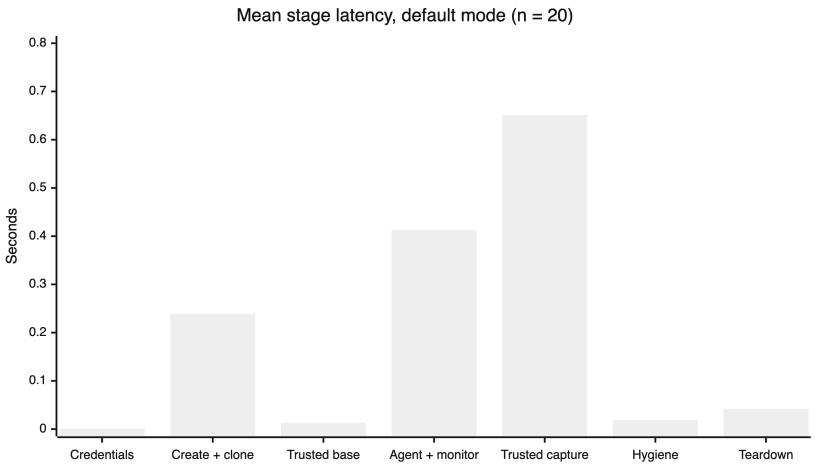



**Figure 7.1:** Mean stage latency in the default configuration

Trusted capture is the most expensive stage because it starts a second container, extracts the work tree and rebuilds the repository; this is the price of a diff the agent cannot falsify. With a real agent, model response time would dominate the agent-execution stage and capture costs would grow with the size of the change, so these stage shares apply only to the mock agent's one-file change. Agent execution took 0.413 s against 0.186 s on the host, the difference covering the in-container watcher and Docker's exec API. Table 7.4 and Figure 7.2 compare the configurations.

**Table 7.4:** End-to-end latency by configuration

| Configuration | Runs | Mean time, s |
|---|---:|---:|
| Direct on host: clone and agent | 20 | 0.237 |
| VaultShield default | 20 | 1.403 |
| VaultShield with egress allowlist | 10 | 2.213 |
| VaultShield with one screenshot route | 10 | 3.294 |

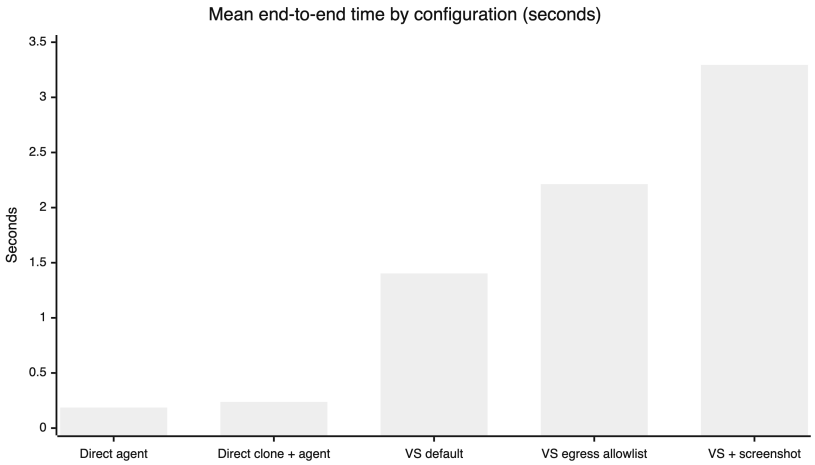



**Figure 7.2:** Mean end-to-end time by configuration

The egress allowlist added 0.810 s, mostly for creating the per-run networks and proxy

and removing them at teardown. One screenshot added 1.891 s, almost all of it spent starting the application and waiting for it to answer.

## 7.4 Overhead with and without VaultShield

Against direct execution with a fresh clone, VaultShield's default mode added 1.166 s per run and the egress allowlist mode added 1.976 s with the mock agent. If this overhead stayed fixed, it would be 0.39% and 0.66% of a five-minute agent session. This is a projection, not a measurement: it assumes that a real agent's session is unaffected by the sandbox and that capture costs do not grow with the change, neither of which has been measured. Table 7.5 places the overhead alongside the qualitative differences.

**Table 7.5:** Running an agent with and without VaultShield

| Dimension | Direct execution | VaultShield |
|---|---|---|
| Fixed overhead | – | +1.166 s, or +1.976 s with egress |
| Host checkout unchanged | No | Yes, in every run |
| Real .env values visible to the agent | Yes | No |
| Change captured on a reviewable branch | No | Yes, in every run |
| Outbound traffic restricted and logged | No | With egress allowlist |
| Approval required before a PR | No | Yes |
| Containers or networks left behind | – | 0 |

## 7.5 Safety Events Captured

Table 7.6 summarises the safety-relevant results across the 40 VaultShield benchmark runs and the security-focused test cases. These are rates over injected scenarios, which show that each control works for the behaviour it was designed against but do not prove the controls are exhaustive.

**Table 7.6:** Safety events and evidence captured

| Scenario | Result |
|---|---|
| Host checkout unchanged after the run | 40 of 40 |
| Agent's file present in diff.json | 40 of 40 |
| File-change event recorded for the agent's file | 32 of 40 |
| Non-allowlisted connections blocked and logged | 4 of 4 |
| Credential-leak vectors detected | 5 of 5 |
| Tampered in-container Git defeated | 2 of 2 |

| Scenario | Result |
|---|---|
| Values in six real .env.sandbox files that are dummies | 30 of 30 |
| Containers and networks left after the campaign | 0 |

## 7.6 Secret Exposure With and Without Protection

The feasibility spike asked a production coding agent to read .env and .env.local and report the exact values of two fake keys. Table 7.7 reports the outcome under four protection configurations.

**Table 7.7:** Secret exposure under four protection configurations

| Config | Protection layers | Verdict |
|---|---|---|
| C0 | None | Both secrets printed |
| C1 | Agent deny rules for .env files | .env.local read into context |
| C2 | C1 plus a hook denying shell access to .env | .env.local read into context |
| C3 | C2 plus real values replaced by a placeholder on disk | Safe |

Agent-level deny rules proved brittle, and only keeping the real value off disk was safe. This is why the sandbox receives dummy values and why the Protect module moves values into the macOS Keychain for agents run outside the sandbox. Unlike the benchmark, this experiment used a real production coding agent, but a single agent and a single prompt, so it shows that deny rules can fail rather than how often they fail.

## 7.7 Case Study: Analysis of Real Runs

The repository preserves five complete mock-agent runs from development. In run 20261009-091725-ff669fcd7dd00b2b, the event log records seven events: the agent confirms that its working directory is the sandbox clone, narrates its work over 157.9 ms, and the in-container watcher then independently reports the created file 106.5 ms after the agent's write. The diff contains exactly that file, and the screenshot shows the sandbox clone's directory listing. In one of the five runs, however, the event log has no file-change event even though the diff and screenshot both show the file, the same gap seen in 8 of the 40 benchmark runs. The event log is therefore corroborating evidence of behaviour, while the trusted diff is the authoritative record of what changed.

### 7.7.1 Execution Evidence from a Live Run

To complement the benchmark tables, the complete system was executed on 1 November 2026 against the running Docker Compose stack and the public repository octocat/Hello-World, using the mock agent with the egress allowlist enabled. The figures below are

captures of those real executions. Terminal output was saved from the commands and rendered in a terminal frame without editing; where an output was long, the figure shows an excerpt of selected lines, marked as such in the figure. The dashboard images are screenshots of the live dashboard, cropped to the run panel.

Figure 7.3 shows the run itself. The run was registered with the backend before the sandbox was created, the dummy credential file was built from the developer's .env, the sandbox was moved onto the egress-isolated network, the change was committed to the run branch in the capture helper, and the sandbox was torn down before the evidence was uploaded. The whole command took 6.89 s of wall-clock time including the clone from GitHub, the backend upload and one screenshot.

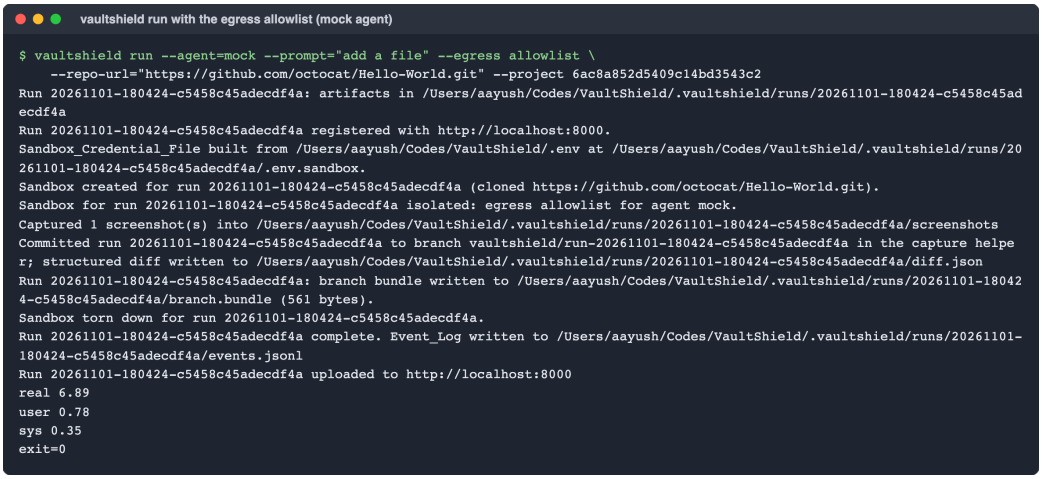



**Figure 7.3:** A real vaultshield run with the egress allowlist and the mock agent

Figure 7.4 shows the artifacts that run produced. Every value in .env.sandbox is a dummy of the same shape as the real value, including a database URL whose credentials and host were replaced, and the event log records the agent's output lines and the watcher's independent file-change event. The structured diff contains exactly the one file the agent created.

Figure 7.5 and Figure 7.6 show the same run in the live dashboard, awaiting review, with its timeline and diff.

Figure 7.7 shows the approval gate and the cleanup check. Submitting the run while it was still awaiting review was refused before any side effect, no VaultShield sandbox, capture helper or proxy container remained, the only remaining network was the Compose stack's own, and the agent's file did not exist in the host checkout.

Figure 7.8 shows the isolation probe, a standalone script that creates a real sandbox through the same class the run command uses and tries what a hostile agent might try. The host .env and the Docker socket were absent, the working directory was the sandbox clone, and only the dummy credential file was readable. The home directory listed by the probe is the container's own /root, which contains only image defaults.

Figure 7.9 shows the Docker-backed security tests run against real containers: a blocked connection is blocked and recorded, DNS exfiltration is blocked, all five token-leak vectors are detected, and a tampered in-container Git cannot hide a file from the branch or diff. All 15 tests passed in 34.4 s.

Artifacts written by the run

```
$ cat .env.sandbox            # dummy values only
VAULTSHIELD_MONGODB_URL=mongodb+srv://dummy_user:dummy_password@localhost/?retryWrites=true&w=majority
VAULTSHIELD_MONGODB_NAME=dummy-value
SECRET_KEY=dummy-value
DEBUG=dummy-value
ALLOWED_HOSTS=dummy-value
GEMINI_API_KEY=dummy-value
GROQ_API_KEY=z6uyz8QLg1ajKg9BC9ZKJAqDIOmvIZOwicsbOb1CGETWdwEKRk8OFdzD

$ cut -c1-150 events.jsonl     # lines cut at 150 characters
{"event_type": "output", "timestamp": "2026-11-01T18:04:26.582646+00:00", "payload": "[mock-sandbox] cwd=/workspace/repo"}
{"event_type": "output", "timestamp": "2026-11-01T18:04:26.582706+00:00", "payload": "[mock-sandbox step=0] simulated output f
or prompt='add a file'"}
{"event_type": "output", "timestamp": "2026-11-01T18:04:26.637693+00:00", "payload": "[mock-sandbox step=1] simulated output f
or prompt='add a file'"}
{"event_type": "output", "timestamp": "2026-11-01T18:04:26.693660+00:00", "payload": "[mock-sandbox step=2] simulated output f
or prompt='add a file'"}
{"event_type": "output", "timestamp": "2026-11-01T18:04:26.750056+00:00", "payload": "[mock-sandbox] wrote vaultshield_mock_ag
ent.txt"}
{"event_type": "output", "timestamp": "2026-11-01T18:04:26.750116+00:00", "payload": "[mock-sandbox] simulated final message f
or prompt='add a file'"}
{"event_type": "file_change", "timestamp": "2026-11-01T18:04:26.862517Z", "payload": "created: /workspace/repo/vaultshield_moc
k_agent.txt"}

# diff.json, pretty-printed with Python's json module, lines cut at 150 characters
{
 "files": [
  {
   "path": "vaultshield_mock_agent.txt",
   "additions": 1,
   "deletions": 0,
   "patch": "diff --git a/vaultshield_mock_agent.txt b/vaultshield_mock_agent.txt\nnew file mode 100644\nindex 0000000..f59c89
5\n--- /dev/null\n+++ b/
  }
 ]
}
```

**Figure 7.4:** Run artifacts: dummy credential file, event log and structured diff

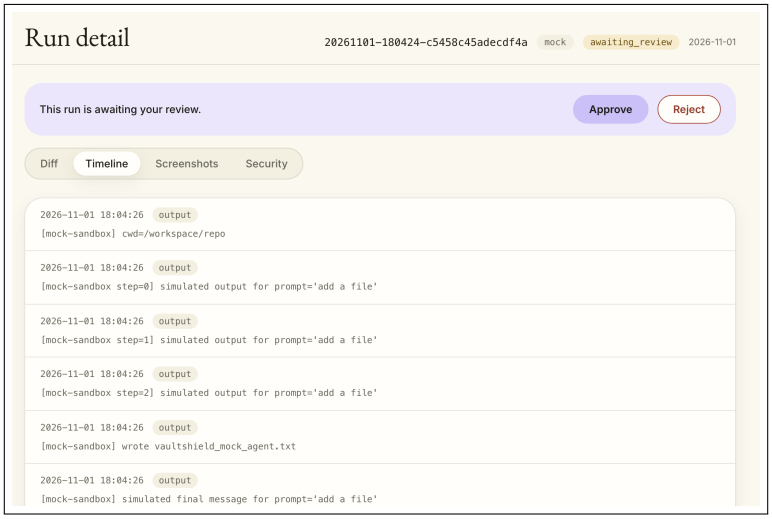



**Figure 7.5:** Live dashboard: Timeline tab of the real run




**Figure 7.6:** Live dashboard: Diff tab of the real run, awaiting review

Approval gate and cleanup check after the run

```
$ vaultshield submit 20261101-180424-c5458c45adecdf4a
Error: Run 20261101-180424-c5458c45adecdf4a is 'awaiting_review', not 'approved', so it cannot be submitted. Nothing was pushed
and no pull request was opened.
exit=1

$ docker ps -a --filter name=vaultshield-run --filter name=vaultshield-capture --filter name=vaultshield-egress
NAMES     STATUS

$ docker network ls --filter name=vaultshield
NAME                 DRIVER
vaultshield_default   bridge

$ ls ../vaultshield_mock_agent.txt   # run from cli/: the host checkout root
ls: ../vaultshield_mock_agent.txt: No such file or directory
```

**Figure 7.7:** Approval gate refusing an unapproved run, and no leftover containers or host changes

isolation_probe.py: hostile-agent probes in a real sandbox

```
$ python isolation_probe.py   # excerpt: probe headings and errors
[+] Creating Sandbox (base_image=python:3.11, repo=https://github.com/octocat/Hello-World.git) ...
[+] Sandbox created. Running probes:
--- Current working directory [OK] ---
/workspace/repo
--- List cwd contents [OK] ---
--- Read host .env (should FAIL) [exit=1] ---
cat: /host/.env: No such file or directory
--- Read mounted .env.sandbox (should WORK) [OK] ---
--- List host home directory (should FAIL) [OK] ---
--- List /home (should be empty/absent) [OK] ---
--- Access Docker socket (should FAIL) [exit=2] ---
ls: cannot access '/var/run/docker.sock': No such file or directory
--- Attempt Docker socket read via curl (should FAIL) [OK] ---
--- Network identity / hostname [OK] ---
--- Write a test file into the repo [OK] ---
[+] Tearing down Sandbox ...
[+] Done.
```

**Figure 7.8:** Isolation probe run inside a real sandbox (excerpt of probe headings and errors)

Docker-backed egress, token-hygiene and tamper tests

```
$ pytest -v tests/test_egress_docker.py   # excerpt: result lines
tests/test_egress_docker.py::test_a_blocked_connection_is_blocked_and_recorded PASSED [  6%]
tests/test_egress_docker.py::test_an_allowlisted_host_is_reachable PASSED [ 13%]
tests/test_egress_docker.py::test_dns_exfil_is_blocked PASSED            [ 20%]
tests/test_egress_docker.py::test_the_app_port_is_reachable_through_the_proxy PASSED [ 26%]
tests/test_egress_docker.py::test_run_with_egress_allowlist_end_to_end PASSED [ 33%]
tests/test_egress_docker.py::test_token_hygiene_vectors_in_a_real_container[copy] PASSED [ 40%]
tests/test_egress_docker.py::test_token_hygiene_vectors_in_a_real_container[core_worktree] PASSED [ 46%]
tests/test_egress_docker.py::test_token_hygiene_vectors_in_a_real_container[symlink] PASSED [ 53%]
tests/test_egress_docker.py::test_token_hygiene_vectors_in_a_real_container[gitignored] PASSED [ 60%]
tests/test_egress_docker.py::test_token_hygiene_vectors_in_a_real_container[tracked_symlink] PASSED [ 66%]
tests/test_egress_docker.py::test_credential_snapshot_never_follows_symlinks_out_of_the_root PASSED [ 73%]
tests/test_egress_docker.py::test_a_tampered_in_container_git_cannot_hide_a_file_from_the_branch_or_diff[True] PASSED [ 80%]
tests/test_egress_docker.py::test_a_tampered_in_container_git_cannot_hide_a_file_from_the_branch_or_diff[False] PASSED [ 86%]
tests/test_egress_docker.py::test_trusted_capture_preserves_non_utf8_and_case_variant_names[A] PASSED [ 93%]
tests/test_egress_docker.py::test_trusted_capture_preserves_non_utf8_and_case_variant_names[B] PASSED [100%]
========================== 15 passed in 34.40s ==========================
```

**Figure 7.9:** Docker-backed egress, token-hygiene and tamper-resistance tests (excerpt of result lines)

## 7.8 Overall Discussion and Limitations

With the mock agent, separating the execution of an agent from trust in its output cost about 1.2 seconds per run, or about 2.0 seconds with network allowlisting. In exchange, every evaluated run left the developer's checkout untouched, kept real secrets out of the agent's environment, recorded an event log, captured the change on a branch the agent could not falsify and held it behind an approval state machine. Table 7.8 lists the main residual risks.

**Table 7.8:** Residual risks and limitations

| Limitation | Consequence | Planned improvement |
|---|---|---|
| Shared-kernel container isolation; default sandbox runs as root | An exploit could reach the host | Default hardening, gVisor or microVMs |
| Egress unrestricted unless enabled | Repository content could be exfiltrated | Default-deny egress |
| Host-based allowlist | Uploads to an allowlisted service are possible | Path-aware filtering |
| Status PATCH can set approved; no multi-user model | A token holder can skip review | Gated transitions, reviewer roles |
| Polling file watcher misses some writes | Timeline may omit a change | Diff cross-check, event-driven watcher |
| Diff tab shows counts, not patch text | Line-level review needs diff.json | Inline patch rendering |

The evaluation itself is limited. All timing figures come from the mock agent on one machine and one small repository, so they describe the pipeline's fixed overhead and must not be generalised to Codex, Kiro, Antigravity or any other model-backed agent. The safety results are rates over injected scenarios and do not measure resistance to an adaptive adversary. Evaluation with real agents is planned for Semester VIII.

### 7.8.1 Comparison with Existing Approaches

Table 7.9 compares the outcome of the same threats under three approaches: running the agent directly on the host, relying on agent-native protection (deny rules, OS sandbox and approval prompts, as documented for Claude Code and Codex CLI [16], [18]), and VaultShield. The direct-execution and VaultShield columns, and the deny-rule result, were measured in this project. The remaining agent-native entries come from vendor documentation and were not measured by us, and "not addressed" means the cited documentation does not describe the control. The product-level comparison of Table 2.2 is not repeated here.

**Table 7.9:** Outcome of the same threats under three approaches

| Threat or property | Direct on host (measured) | Agent-native protection | VaultShield (measured) |
|---|---|---|---|
| Agent reads a secret in the project .env | Both secrets printed (C0) | Deny rules bypassed via .env.local (C1, C2, measured) | No real value present; 30 of 30 values dummies |

| Threat or property | Direct on host (measured) | Agent-native protection | VaultShield (measured) |
|---|---|---|---|
| Agent modifies the developer's checkout | Yes | Yes, edits land in the workspace by design | Host unchanged in 40 of 40 runs |
| Outbound connection to an unlisted host | Allowed | Blocked by default in Codex CLI; configurable in Claude Code | Blocked and logged, 4 of 4, when the allowlist is on |
| Agent hides a change by tampering with Git | Possible | Not addressed in cited documentation | Defeated, 2 of 2 |
| Agent's credential leaks into its output | Not checked | Not addressed in cited documentation | Detected, 5 of 5 vectors |
| Change reaches the repository without review | Yes, if the developer commits | Per-action prompts; no held change | Refused until approved (Fig. 7.7) |
| Works with several agents | – | No, built into one agent | Yes, four adapters |
| Fixed time cost per run (mock agent) | 0.237 s | Not measured | +1.166 s, or +1.976 s with egress |

The comparison shows where VaultShield adds protection that the other approaches do not provide, and also that agent-native sandboxes already block the network by default where VaultShield requires opting in. The two kinds of control are complementary: an agent's own sandbox can still run inside a VaultShield sandbox.

## 7.9 Chapter Summary

With the mock agent, the default pipeline completed in 1.403 s on average, adding 1.166 s to direct execution, with trusted capture the largest stage; real-agent timings remain to be measured. Isolation, change capture and cleanup succeeded in every evaluated run, all injected unsafe behaviours were blocked or detected, and every value in six real sandbox credential files was a dummy. The file watcher recorded the agent's change in 80% of runs, confirming the trusted diff as the authoritative record.

# Chapter 8

# Conclusion

VaultShield separates running an AI coding agent from trusting what it produces. The agent receives a disposable Docker container, a fresh clone and only dummy credentials; its output, file changes and network decisions are recorded in an event log; its change is captured as one commit on a dedicated branch by a separate hardened container; and the change can become a pull or merge request only after a human approves it in the review dashboard. The system was built as a Python CLI, a Django REST Framework backend on MongoDB with Celery workers, and a React dashboard, totalling about 18.1 thousand logical lines of product code. Table 8.1 assesses the outcome against each objective.

**Table 8.1:** Achievement of objectives

| Objective | Evidence | Outcome |
|---|---|---|
| 1. Universal agent execution platform | Mock, codex, kiro and antigravity agents run behind one adapter interface | Achieved |
| 2. Agentic AI Orchestrator | Not implemented in Semester VII; proposed design in Section 3.4.5 | Planned for Semester VIII |
| 3. Secure sandboxed execution | Host checkout and .env unchanged in 40 of 40 runs; dummy credentials and one-shot clone token | Achieved |
| 4. Monitoring and security analysis | Event log and diff for every run; violations recorded; Celery security scan | Achieved, with a file-watcher gap |
| 5. Governance and human approval | Dashboard review and decision controls; submit refuses non-approved runs | Achieved |

Objectives 1, 3, 4 and 5 were achieved in Semester VII to the extent supported by the evaluation, which used a mock agent for all timings. Objective 2, the Agentic AI Orchestrator, was not implemented in Semester VII and remains planned for Semester VIII; its proposed design, workflow and security considerations are given in Chapter 3 (Section 3.4.5). Known limitations remain: the shared-kernel container boundary and opt-in egress control (Section 3.4.6), and two further gaps: in the single-developer MVP a token holder can bypass the dashboard through an unrestricted status update, and the polling file watcher missed some writes in very short runs, although the trusted diff captured every change. The project contributes a working implementation of the governance practices recommended for agentic systems [9], [10], engineering techniques such as shape-preserving credential substitution, a one-shot clone credential and trusted capture, and evidence that the pipeline's own fixed cost is about 1.2 seconds per run with a mock agent, while agent-level deny rules alone do not protect secrets. Real-agent performance has not yet been measured.

# Chapter 9

# Future Scope

The limitations identified in this report define the future work. The first four items are the committed Semester VIII enhancements, each tied to a limitation found in the evaluation; the remaining items are longer-term work.

**Semester VIII enhancements**

1. **Agentic AI Orchestrator:** implement the design in Section 3.4.5, which scans the project, asks clarifying questions before execution and passes a sanitised, enriched prompt to the agent, then measure on real agents whether it raises task success and approval rates and reduces hallucinated files, functions and packages. This completes Objective 2.
2. **Stronger Sandbox Isolation:** run the sandbox by default as a non-root user with all capabilities dropped, no-new-privileges and CPU, memory and process limits; make egress default-deny; and support gVisor or Firecracker microVM runtimes [12], [13]. This addresses the shared-kernel and opt-in egress limitations (threats T3, T7 and T8 in Table 3.3).
3. **Improved Monitoring:** replace the polling file watcher, which missed 8 of 40 short writes, with an event-driven watcher; cross-check the event log against the trusted diff; stream events live so a reviewer can interrupt a run [9]; render patch text in the Diff tab; and enforce the interactive sign-in deadline with a watchdog (Section 6.3.2).
4. **Role-Based Approval:** add separate developer and reviewer accounts, role-based access control, decision attribution and optional multiple approvals, and route every status transition through the gated approval endpoint, so that the approval gate becomes a separation-of-duties control rather than a single-user workflow control (threat T9).

**Further long-term work**

- **Policy Engine:** express rules such as required approvals for infrastructure files or approved dependency lists declaratively, with path-aware egress filtering.
- **Cost and Resource Controls:** record per-run token usage and enforce per-project budgets and sandbox time limits.
- **Multi-Repository and Parallel Runs:** capture one branch per repository in a single run, and let several agents attempt the same task for the reviewer to compare.
- **IDE Integration and Evidence Quality:** start and review runs from the editor, and scan screenshots for credentials.
- **Broader Evaluation:** extend the real-agent evaluation to a subset of SWE-bench [2] and to adaptive red-team attacks, with a usability study with reviewers.

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